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Energy-efficient deep learning-based intrusion detection system for edge computing: a novel DNN-KDQ model

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Abstract

The proliferation of the Internet of Things (IoT) and edge computing technologies has expanded the attack surface for cybercriminals, emphasizing the need for robust and efficient cybersecurity measures. Intrusion detection systems (IDS) are essential defenses; however, deploying traditional IDS solutions on edge devices remains challenging due to limited computational, memory, and energy resources. This research proposes an energy-efficient IDS framework based on a modified Deep Neural Network with Knowledge Distillation and Quantization (DNN-KDQ) to address these challenges. The CICIDS2017 dataset was preprocessed to extract energy-centric features, and adaptive sampling and model compression techniques were applied to optimize performance. The proposed DNN-KDQ model achieves a prediction test accuracy of 99.43%, reduces model size from 196.77 KB to 20.18 KB, and achieves an inference time of 0.07 ms per sample in real-time scenarios. These results demonstrate the feasibility of deploying high-accuracy, low-latency IDS models on resource-constrained edge devices, contributing to more scalable and energy-efficient cybersecurity solutions for modern network infrastructures.

Keywords DNN-KDQ model, Intrusion detection system (IDS), Deep learning models, Energy efficiency, Cybersecurity, Edge computing, Model compression, Knowledge distillation, Quantized, Real-Time anomaly detection

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Introduction

Context

The digitization process provides numerous benefits to the energy sector, including improved forecasting, predictive maintenance, demand-side opportunities, enhanced system operation, and more efficient measurement and monitoring systems [1]. Additionally, digital transformation has contributed to mitigating cybersecurity challenges [2]. However, the energy industry remains highly vulnerable to cyber threats such as supply chain attacks, ransomware, and data breaches. The increasing reliance on digital technologies and widespread data usage necessitates strict adherence to privacy, security, safety, and ethical standards, particularly concerning cybersecurity threats [2].

The scalability and operational efficiency of energy systems have significantly improved with advancements in digitalization, including smart grids [1, 3] renewable energy networks [4], and Internet of Things (IoT)-based energy management devices [2, 5]. Despite these improvements, digital transformation introduces new cybersecurity risks. Cyberattacks targeting these infrastructures, such as Distributed Denial of Service (DDoS), unauthorized access, and data manipulation, can cause operational disruptions, financial losses, and security breaches [6]. Strong cybersecurity measures, including network segmentation, intrusion detection systems (IDS), and regular security audits, can help mitigate these risks [7]. Furthermore, employee awareness and training programs are crucial for recognizing common attack vectors and identifying suspicious activities.

IDS play a vital role in protecting energy infrastructures by monitoring network traffic and detecting potential security threats. These systems generate alerts upon detecting unauthorized access or suspicious activity. IDS solutions can be categorized into three main types: signature-based, anomaly-based, and hybrid IDSs. A primary challenge for signature-based IDS is the need for continuous updates to detect emerging threats effectively [8]. Traditional IDS techniques, such as rule-based and signature-based methods, have shown effectiveness against known threats [9]. However, deploying deep learning (DL)-based IDS solutions in energy systems presents challenges, including high computational requirements, energy inefficiency, and latency concerns. These issues are critical in real-time and resource-constrained operational environments.

Traditional IDS models, while effective in centralized or high-performance computing environments, encounter significant challenges when deployed in edge environments. Specifically, these models typically demand substantial computational power, large memory

footprints, and persistent network connectivity, all of which are constrained in edge devices. Edge computing environments often involve battery-powered or low-resource devices that prioritize energy efficiency, making heavyweight IDS models impractical. Moreover, traditional IDS solutions are not optimized for real-time processing under tight latency constraints, which is critical for timely threat detection at the edge. The large model sizes and complex feature extraction processes also strain local storage and processing capabilities, resulting in increased inference delays and reduced system responsiveness. These limitations necessitate the development of lightweight, energy-efficient IDS models capable of maintaining high detection accuracy with minimal resource consumption, as addressed in this research.

This research tackles these challenges by developing a DL-based, energy-efficient IDS for modern energy infrastructures. The primary goal is to enhance security without compromising efficiency by designing IDS models optimized for low power consumption and real-time responsiveness. However, DL models often require a large number of parameters, making them computationally expensive. While advancements in hardware, such as high-performance GPUs, TPUs, and cloud computing, have facilitated DL adoption, edge devices still face limitations in storage capacity and model inference speed. Researchers have explored lightweight DL techniques, such as quantization, knowledge distillation, and filter pruning, to address these constraints [10, 11].

Weight compression techniques aim to reduce the memory requirements of model parameters, often leveraging efficient encoding strategies to minimize storage and memory access demands. Quantization simplifies computational processes, reduces model size, and optimizes bit-width for local computations on IoT devices. By replacing complex 32-bit floating-point calculations with lower-bit multiplications, inference acceleration is achieved. Low-precision quantization techniques transform full-precision weights and activations into 8-bit integer values using integer-arithmetic inference frameworks, while maintaining bias parameters at 32-bit to preserve baseline accuracy [12].

Various intrusion detection techniques and analytical models are available for securing energy systems. DL, a subset of machine learning, leverages multi-layered artificial neural networks to learn hierarchical data representations. Its ability to generate feature hierarchies from large-scale unlabeled data makes it well-suited for analyzing complex, high-dimensional datasets [13]. Many IDS solutions incorporate DL models, achieving high detection performance. Figure 1 provides an overview of the real-time DL-Based IDS Approach.

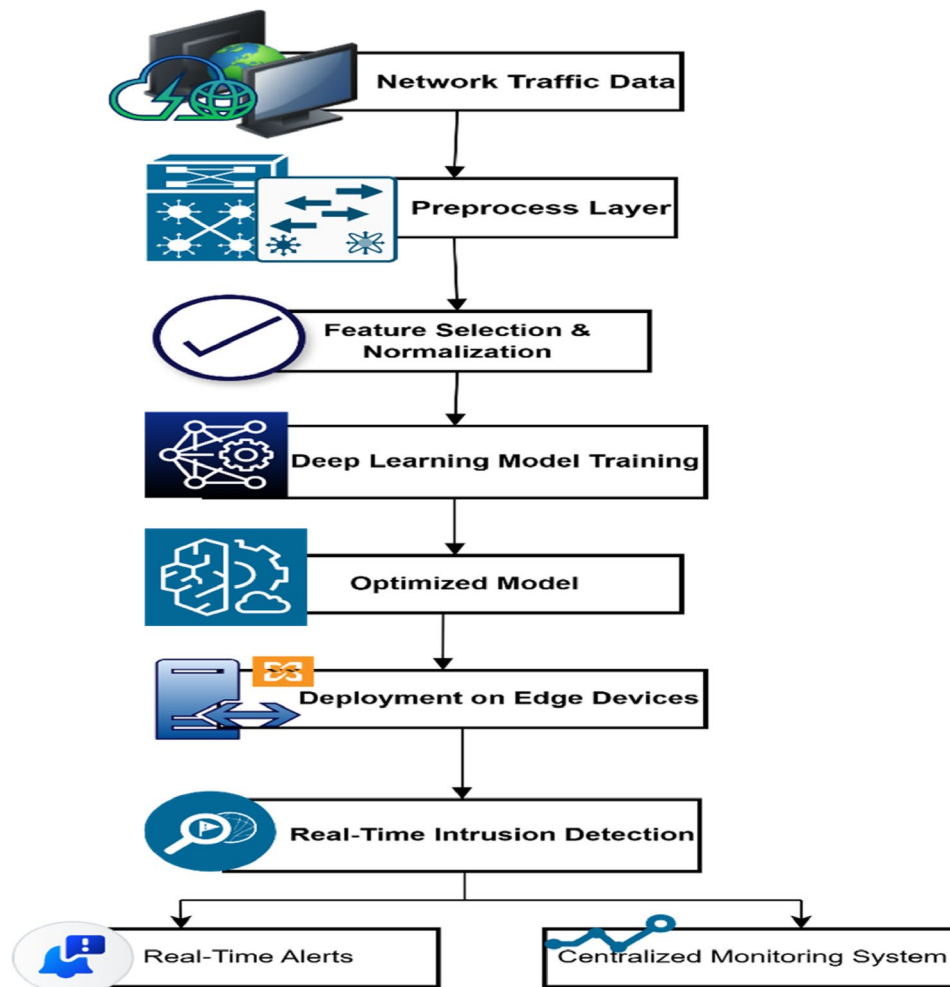


Fig. 1 A DL-Based IDS Approach

Objective

With the increasing interconnectivity of energy systems, they have become more susceptible to cyberattacks. One of the critical challenges in these systems is the need for real-time threat detection, yet existing IDS often experience delays due to high computational demands. Many energy systems, particularly edge and IoT devices, have limited energy and processing capabilities, making them unsuitable for complex IDS solutions. While DL can enhance intrusion detection, its high resource requirements pose significant constraints. The primary objective is to design an IDS that achieves an optimal balance between energy efficiency, detection accuracy, and real-time performance. To address these challenges, this research introduces the DNN-KDQ model, which leverages DL and energy optimization techniques to enhance security in energy systems.

Key aspects and novel elements

The main contributions may be summarized as follows:

- Development of a lightweight, energy-efficient IDS specifically tailored for edge computing environments, overcoming traditional IDS limitations such as high computational demands and latency issues
- Design and implementation of the DNN-KDQ model that integrates deep learning with knowledge distillation and quantization techniques, significantly reducing model size (from 196.77 KB to 20.18 KB) and improving inference time (0.19 ms per sample).
- Comprehensive preprocessing of the CICIDS2017 dataset using an energy-centric approach, including feature selection, correlation analysis, and stratified sampling to maximize detection performance while minimizing computational overhead.
- Comparison and evaluation of different deep learning architectures (DNN, CNN, RNN, CNN-LSTM) to validate the superior energy efficiency and accuracy of the proposed DNN-KDQ model.
- Real-time deployment of the DNN-KDQ model on edge devices, demonstrating high performance with

a prediction test accuracy of 99.43% and a real-time inference time of 0.07 ms.

- Integration of adaptive sampling and sliding window techniques to further enhance energy savings and ensure consistent real-time intrusion detection without sacrificing accuracy.
- Proposition of an energy-optimized cybersecurity framework that combines anomaly detection, real-time alerting, and automated mitigation strategies for modern energy infrastructures.

The novelty of the proposed DNN-KDQ model lies in its unique combination of deep learning, knowledge distillation, and quantization specifically adapted for energy-constrained edge computing environments. Unlike existing IDS solutions, which often focus either on detection accuracy or on lightweight deployment separately, the DNN-KDQ model simultaneously optimizes for high intrusion detection performance, drastic model size reduction, ultra-low inference time, and real-time operation on edge devices. Furthermore, the integration of adaptive sampling, sliding window techniques, and energy-centric feature preprocessing differentiates the proposed IDS from traditional and current lightweight models by providing a holistic optimization framework suitable for modern IoT and smart energy infrastructures.

Structure of the article

The remainder of this paper is structured as follows: Sect. 2 presents a review of the relevant literature and previous work on the topic. Section 3 introduces the proposed technique, detailing its methodology and implementation. Section 4 discusses the experimental results and evaluation metrics, concluding with a comprehensive discussion. Finally, Sect. 5 provides the conclusion and outlines potential future research directions.

Literature review

Current research

IDS play a crucial role in securing IoT networks against cyber threats. Various approaches have been proposed to enhance the performance, accuracy, and efficiency of IDS solutions. This section reviews and discusses the diverse literature on ML-based IDS, highlighting their strengths and limitations.

Olanrewaju-George and Pranggono [14] developed an IDS leveraging federated learning to enhance anomaly detection and data privacy in IoT devices. Both supervised and unsupervised DL models were trained, demonstrating improved performance compared to non-federated learning methods. However, the approach

lacks an evaluation of anomaly patterns across different devices, limiting its generalizability.

Prazeres et al. [15] proposed a distributed IDS architecture utilizing flow-based anomaly detection optimized for resource consumption in IoT environments. The approach employed deep neural networks to detect malicious traffic and demonstrated a strong performance with 0.2 false positives and 0.1 false negatives. However, challenges remain in feature selection for anomaly detection due to device variability.

Poornima et al. [16] introduced a hybrid LDA-LR-based IDS for edge computing, achieving an accuracy of 96.56% and a precision of 95.78%. The approach efficiently detected various attack types but lacked scalability analysis concerning the increasing number of IoT devices. Additionally, privacy issues in IoT data transfer remain an open research challenge.

Adabi et al. [17] integrated an IDS with a Security Information and Event Management (SIEM) system to provide real-time cyberattack detection. The system effectively identified DoS attacks, but high CPU and RAM usage in Elasticsearch indicated the need for optimization to reduce resource consumption.

Qing et al. [18] introduced a robust multi-stage progressive autoencoder for anomaly detection in hyperspectral images. This approach demonstrated superior detection capabilities compared to state-of-the-art methods. However, anomalies remain challenging to detect due to low occurrence probabilities, and feature extraction techniques require further improvements for better detection accuracy.

Shahid et al. [19] proposed an optimized IDS for Industrial IoT (IIoT) networks using Deep Transfer Learning (DTL) and bootstrap aggregation ensemble techniques. The model achieved 100% accuracy across 14 cyberattack classes. However, the study did not address real-time intrusion detection challenges and lacked extensive testing on diverse IIoT environments.

The Internet of Things (IoT) is a relatively recent domain in information technology that enables any object to communicate with any other device via a network. In recent years, cloud computing, big data, industrial wireless networks (IWNs), and the Industrial Internet of Things (IIoT) have significantly advanced. For IIoT to function, it requires a dependable and efficient data collection system, akin to a spanning tree. Early spanning tree algorithms disregarded failure and mobility entirely [20].

By replacing “Things” with “Drones,” the Internet of Drones (IoD) surpasses the Internet of Things (IoT) while retaining the unique characteristics of both ideas. IoD technologies have garnered a lot of attention lately

because of their many useful applications, but convincing people that they are safe is still difficult. In the Internet of Devices (IoD), intrusion detection systems (IDSs) find it difficult to adjust to the constantly changing network architecture. Finding the ideal balance between detection speed and accuracy is one of the biggest obstacles. The use of radial basis function neural networks (RBFNNs) to improve performance is illustrated in this paper [21].

The Internet of Drones (IoD) maintains the particular qualities distinguishing the two concepts by substituting “Drones” for “Things,” therefore replacing the Internet of Things (IoT). IoD technologies have drawn much interest recently because of their many significant and diverse applications. It is hard to reach the required degree of public acceptance since IoD is not easy to show as safe and secure for human life. Furthermore, because network architecture is constantly evolving, intrusion detection systems (IDSs) in the Internet of Devices (IoD) struggle greatly. One of the key challenges is finding the perfect balance between the speed of threat detection and accuracy. This paper [22] demonstrates how to use radial basis function neural networks (RBFNNs) to enhance the performance of the Internet of Devices (IoD) network using blockchain technology.

Botnets, which are networks of compromised computers controlled by command and control (C&C) servers, pose a serious threat to internet security. Phishing, spam distribution, and denial of service (DoS) attacks are some of the destructive actions enabled by these malevolent forces. Every bot operates as a malicious binary code that grants attackers remote access on compromised hosts. The attackers are able to execute coordinated, extremely lethal attacks while staying anonymous by utilizing the combined processing power of compromised hosts [23].

K. Sedhuramalingam et al. [24] improved IDS effectiveness in wireless sensor networks by reducing false alarm rates through DL models. The proposed IDS achieved an accuracy of 95%, a precision ratio of 94%, and an F1-score of 95%. However, high computational costs limited extensive experimental testing, highlighting the need for lightweight IDS models in resource-constrained environments.

Kaushik et al. [25] introduced a lightweight IDS using statistical feature selection techniques, which significantly improved detection accuracy while reducing computational overhead. The study demonstrated that machine learning models, particularly Random Forest and AdaBoost, outperform traditional algorithms for IDS. The proposed approach is relevant for optimizing feature selection and minimizing resource consumption in energy-efficient IDS solutions.

Ali et al. [7] addressed intrusion detection in Industrial Control Systems (ICS) networks by developing an IDS framework capable of handling imbalanced datasets. The model employed big data analysis techniques to improve detection accuracy, achieving 99% accuracy on an industrial network dataset. This work is significant for enhancing IDS performance in critical infrastructure security, providing insights into handling large-scale cyberattack detection in IoT environments.

Similarly, recent work has explored the convergence of generative AI, IoT, and blockchain to enhance data security, efficiency, and scalability in healthcare IoT environments, emphasizing the need for integrated and resource-conscious AI deployments [26].

The growing adoption of DL-based IDS necessitates further research into optimizing energy efficiency, reducing computational complexity, and ensuring scalability across diverse IoT environments. The insights from these studies inform the development of more effective, lightweight IDS frameworks tailored for real-time applications in edge computing.

This paper's [27] discusses three fundamental algorithms. The study [27] demonstrates how to use the MapReduce framework in conjunction with Grey Wolf Optimization (GWO) to create services that offer higher quality. With the proposed approach, energy savings increase by 40%, availability increases by 11%, response time decreases by 14%, and cost decreases by 24%.

The fast rise in message rebroadcasts suggests broadcast storms during data transmission in vehicular ad hoc networks (VANETs). Node mobility, regular topology changes, and ongoing network disturbances hinder attempts to create effective broadcasting protocols. Based on a novel fuzzy technique called multicriteria decision-making (MCDM), the study [28] offers a creative way for cars to give priority to ideal neighbors for data broadcasting.

A charge management system based on machine learning (ML) guides electric vehicles (EVs) to charging stations considering traditional charging, fast charging, and vehicle-to-grid (V2G) technologies. This function reduces the expenses related to high voltages, load variations, charging, and power loss. The study [29] evaluate the effectiveness of several machine learning (ML) methods. These approaches include long short-term memory (LSTM), random forests (RF), support vector machines (SVM), decision trees (DT), K-nearest neighbors (KNN), and neural networks (DNN). The results imply that in some circumstances LSTM could be beneficial for electric vehicle control. Reducing the load curve might help the LSTM model's peak voltage, voltage stability, and power losses. We also work to keep our charges low.

Research gap

Despite significant advances in the development of machine learning and deep learning-based IDS for IoT environments, several critical challenges remain unresolved.

First, many existing IDS models suffer from high computational complexity and energy consumption, which makes them unsuitable for deployment on resource-constrained edge devices. Although some studies have proposed lightweight frameworks, few have systematically optimized both computational overhead and energy efficiency while maintaining high detection accuracy, particularly under real-time constraints.

Second, while model compression techniques such as knowledge distillation and quantization have been explored extensively in other domains (e.g., natural language processing, computer vision), their application in IDS for IoT and edge environments remains limited. Few studies have integrated these compression strategies specifically to balance energy efficiency, detection performance, and inference speed in intrusion detection tasks.

Third, most IDS solutions have not been rigorously adapted for scalable, real-time operation across dynamic network conditions, especially when considering diverse attack patterns and varying traffic loads. Real-time adaptability, energy-aware sampling, and edge deployment strategies are often underexplored in traditional IDS research.

The proposed DNN-KDQ model directly addresses these research gaps by introducing an energy-optimized deep learning framework that combines knowledge

distillation, post-training quantization, and adaptive sampling for efficient, real-time intrusion detection on edge devices. This holistic approach not only reduces model size and inference latency but also enhances detection robustness across varying network conditions, paving the way for practical and scalable deployment in modern IoT ecosystems.

Methodology

This study aims to develop a DL-based secure IDS that ensures reliable and energy-efficient cybersecurity for contemporary network infrastructures. Figure 2, represents the framework of our proposed methodology.

Dataset selection and preprocessing

Selecting minimal but relevant features reduces computation and energy usage during model training and inference. In our research, the CICIDS2017 dataset is utilized for training and evaluation. It includes a comprehensive range of benign and attack traffic, including:

- DOS attacks (GoldenEye, Hulk, SlowHTTPTest).
- Brute Force attacks (FTP-Patator, SSH-Patator).
- Web Attacks (Brute Force, SQL Injection).

A well-known benchmark for IDS, the CICIDS2017 dataset was selected because it provides a comprehensive representation of various attack methods and realistic network circumstances. Based on the energy-centric approach, we used a few key features, such as traffic-flow-based features, packet-based features, time-based

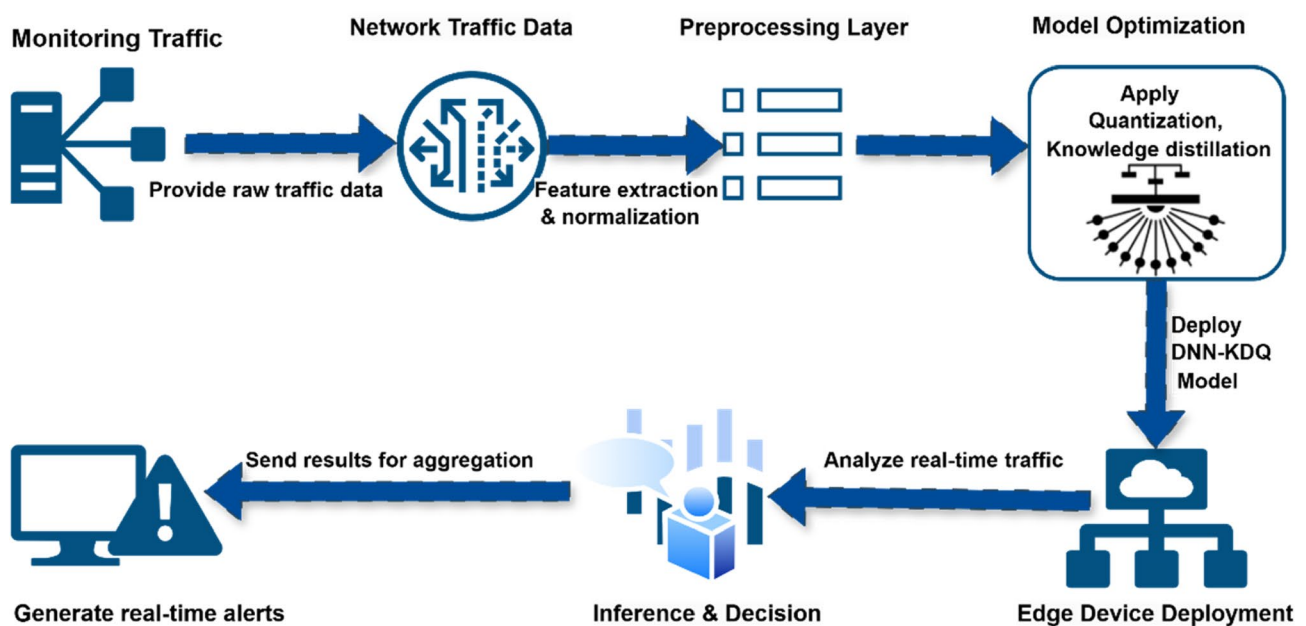


Fig. 2 Real-time intrusion detection on edge devices: a low-power DL framework

features, flag counts, TCP window-based features, active and idle times, and class labels, to maximize the model's performance while reducing computational overhead. To balance detection performance and energy usage, dynamically modify sample sizes according to traffic load. Figure 3 shows histograms for multiple feature distribution. Each subplot corresponds to a different feature, and the x-axis and y-axis illustrate each feature's range and frequency of values. The histogram for Label likely indicates class distributions. The classes may be imbalanced if the dataset is for classification (e.g., benign vs. malicious traffic).

Every experiment was conducted using Ubuntu 20.04 LTS on a workstation equipped with an Intel Core i9-11900 K CPU (3.5 GHz, 8 cores, 16 threads), 64 GB of RAM, and an NVIDIA RTX 3090 GPU (24 GB VRAM). Python 3.8 was used for model development and training, with TensorFlow 2.10.0 and Keras serving as the

foundational frameworks. These were supported by the Scikit-learn 1.1.1, NumPy 1.21.0, and Pandas 1.4.2 libraries. TensorFlow Lite was used for quantization and optimization in order to deploy in systems with restricted resources. For edge deployment testing, a Raspberry Pi 4 (ARM Cortex-A72 CPU, 4 GB RAM) was used to replicate actual IoT environments. On both high-performance and edge systems, the architecture enables effective training, quantization, and real-time testing of the proposed DNN-KDQ model.

The selection of the CICIDS2017 dataset is particularly well-suited for energy-efficient IDS research because it offers a comprehensive and realistic representation of modern network traffic while providing a rich set of features critical for selective feature extraction. Unlike older datasets such as KDD99 or DARPA98, which lack traffic diversity and realistic payloads [30], CICIDS2017 includes up-to-date benign and attack flows captured

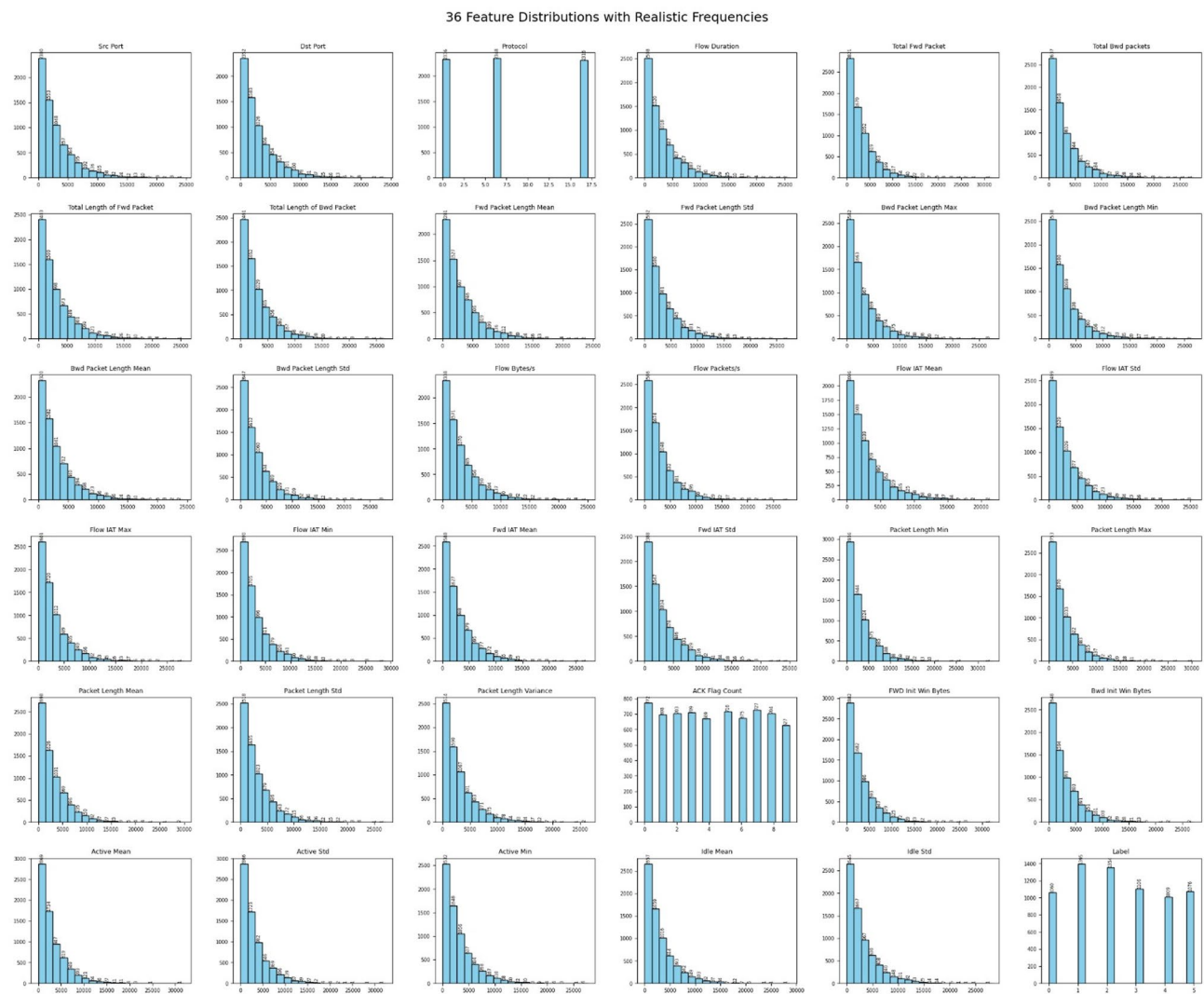


Fig. 3 Feature Distribution of the CICIDS2017 dataset

under contemporary network conditions. This enables the application of energy-centric approaches such as minimal feature selection and adaptive sampling without sacrificing detection effectiveness. Furthermore, the dataset's structure supports the identification of highly correlated and energy-relevant features, allowing for the development of lightweight, computationally efficient models optimized for edge deployment. Its public availability and standardized format also enhance reproducibility and comparative evaluation in energy-constrained IDS studies.

After selecting the important energy-centric feature we used data cleaning techniques and Non-numeric fields i.e. Flow ID, Src IP, Dst IP, Timestamp, etc. were removed. Categorical fields like Protocol were encoded using label encoding to make them suitable for machine learning models. Additionally, we employed data normalization to guarantee consistent feature scaling. The Standard-Scaler was utilized to normalize all numerical features, converting the data into a standard normal distribution. Stratified sampling [31] was then used to preserve the proportion of attack classes by splitting the dataset into 80% training and 20% testing data. We analyze the important features using correlation and feature importance analysis techniques [32]. For correlation analysis, Heatmaps were used to visualize feature correlations, identifying redundancies and highly correlated features. Figures 3 and 4 represents the importance of feature correlation.

On the base of important feature correlation, our model selects the parameter for real-time IDS detection.

A Random Forest model was employed to assess and rank the significance of each feature. To enhance intrusion detection accuracy while addressing dimensionality challenges, feature selection techniques were integrated into the Random Forest (RF) approach. This method allows devices to collaborate on intrusion detection while maintaining data privacy by preventing the disclosure of sensitive information [33]. Figure 5 illustrates the key features of our dataset, which were classified using Random Forest techniques.

Model development

By identifying patterns in the network traffic, the DL models can classify an action as either normal or abnormal. To evaluate the effectiveness of DL approaches, we developed and tested different DL models i.e.

Deep neural network (DNN) model

A robust security model is essential for an effective IDS system to enhance network protection. Reducing the number of layers in the Deep Neural Network (DNN) model helps minimize detection time delays and false positives while strengthening defense against cyber threats [34, 35]. The proposed detection strategies are tailored for large-scale networks while ensuring a streamlined design to avoid unnecessary complexity. The input

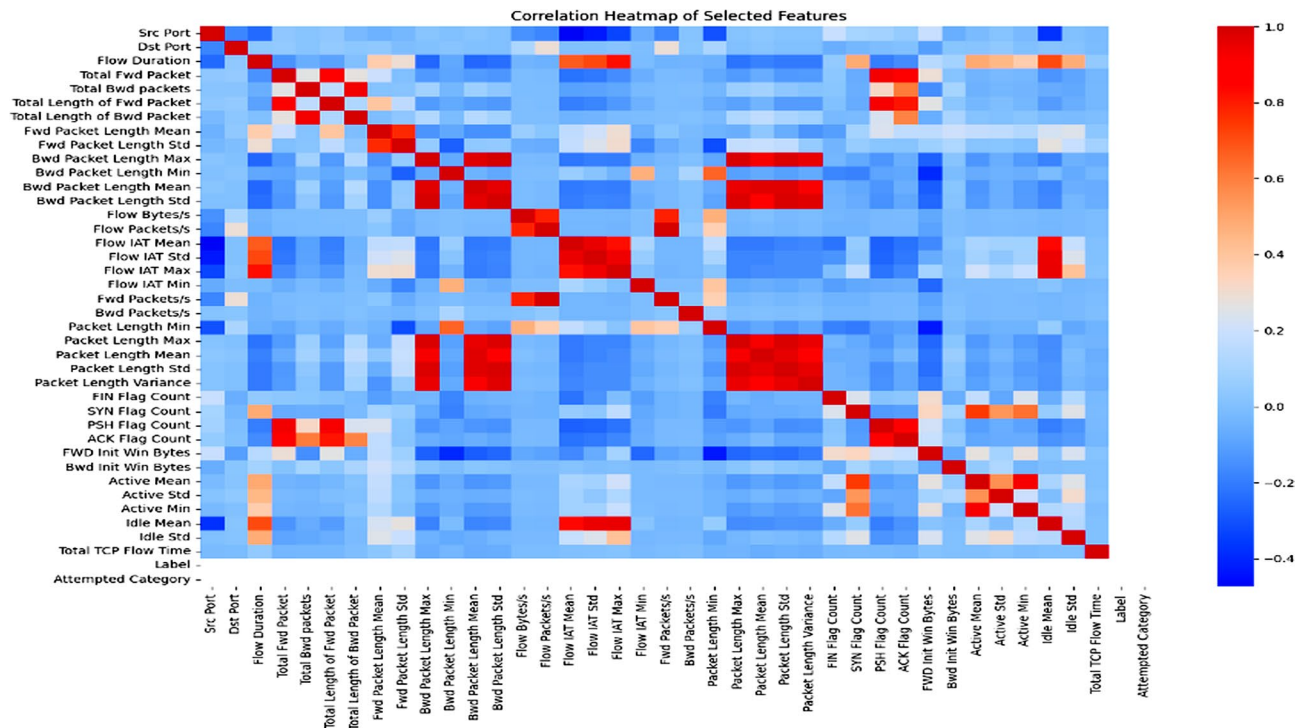


Fig. 4 Important Feature Correlation Analysis

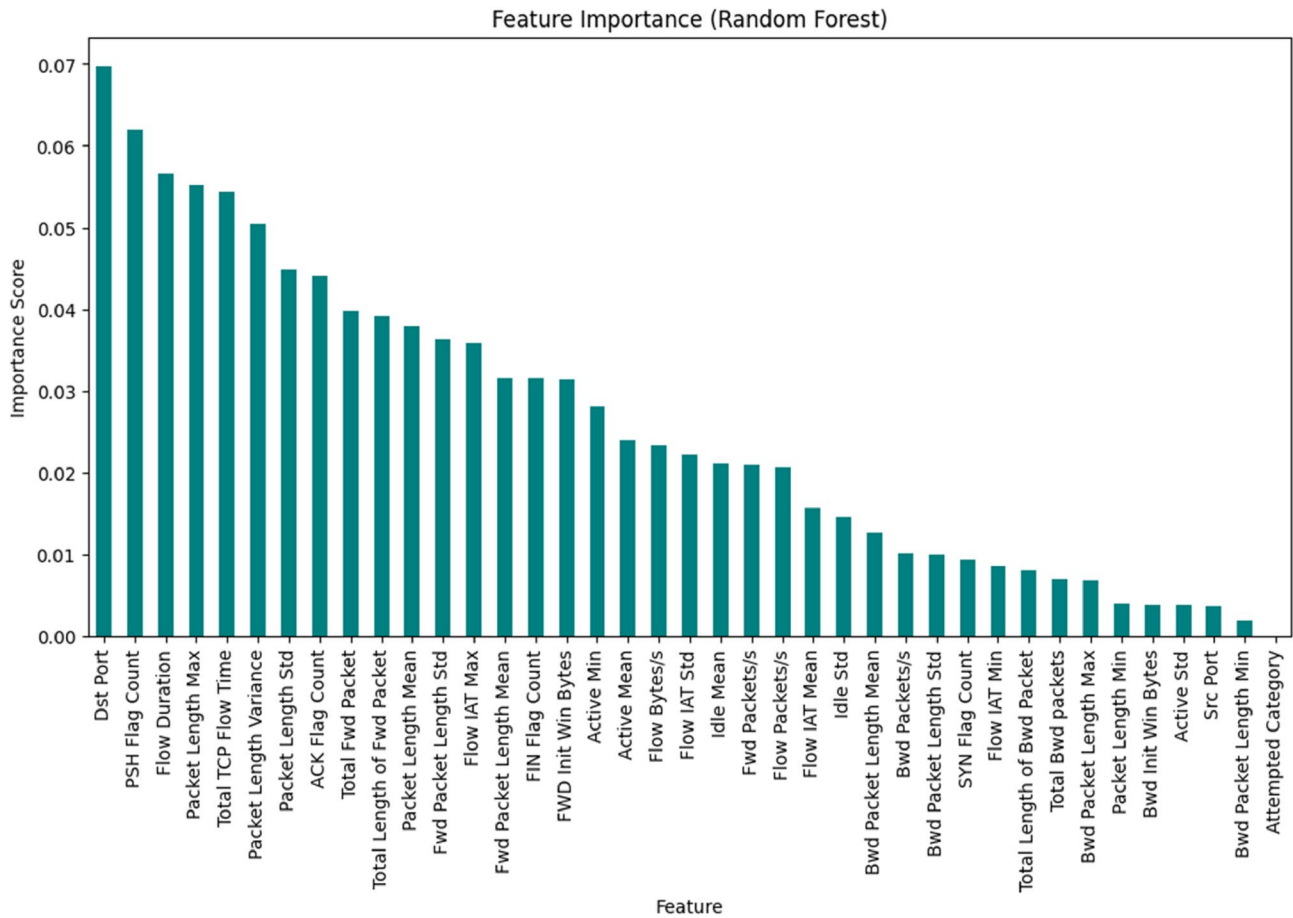


Fig. 5 Analysis Importance Score of features

layer of the sequential model incorporates fully connected layers corresponding to the selected features [36]. To mitigate overfitting, dropout layers are strategically placed after the hidden layers, utilizing the “ReLU” activation function [37]. The optimization process is handled by the “Adam” optimizer, configured with a learning rate of 0.001 and sparse categorical cross-entropy loss. The final output layer classifies intrusion types using a softmax function, ensuring precise detection and categorization of threats.

Convolutional neural network (CNN)

CNNs have long demonstrated their effectiveness in the field of IDS. However, as cyber threats become increasingly sophisticated, the demand for more advanced detection methods continues to rise. CNNs excel at capturing spatial correlations within data, making them well-suited for hierarchical feature extraction [38]. The CNN model leverages convolutional layers with a kernel size of 3×3 , max pooling

layers to reduce feature dimensions, and fully connected layers for classification, effectively transforming input features into 2D matrices. By employing these techniques, CNN architectures enhance their discriminative capacity, reducing noise and focusing on critical information, ultimately improving the accuracy and reliability of IDS detection [39].

Recurrent neural network (RNN)

The RNN model is specifically designed for sequential data analysis. It employs Long Short-Term Memory (LSTM) layers to capture temporal dependencies between network packets, enhancing intrusion detection accuracy [40]. Attack-related features are systematically categorized using the RNN, allowing for effective classification of different types of threats. At each level, the RNN processes incoming inputs while incorporating information from previous computations, ensuring a more comprehensive and context-aware analysis of network activity [41].

Hybrid CNN-LSTM

The hybrid CNN-LSTM model effectively mitigates overfitting and computational complexity while ensuring enhanced scalability by leveraging CNNs for efficient feature extraction and LSTM networks for capturing temporal patterns [42]. Beyond improving detection accuracy, this model addresses the limitations of conventional IDS techniques, providing a more adaptive and robust solution for modern network security challenges. The CNN-LSTM hybrid model integrates LSTM layers for sequential data learning and CNN layers for extracting essential features from the input data, resulting in a highly efficient and precise intrusion detection system.

Proposed DNN-KDQ model

In this research paper, we propose a modified DNN model that integrates energy optimization techniques. We evaluate and compare different deep learning models, including DNN, CNN, RNN, and hybrid CNN-LSTM architectures, based on their effectiveness and energy efficiency. Among these, the DNN model demonstrated superior accuracy and energy efficiency, making it the most suitable choice for implementation

in energy-critical infrastructures with scalability and sustainability considerations.

To further enhance its efficiency, we modified the DNN model by incorporating an energy compression technique. The DNN model serves as the teacher model, from which the student model distills knowledge as needed. Subsequently, we applied quantization techniques to convert 32-bit floating-point numbers into 16-bit or 8-bit integer representations. Utilizing smaller data types accelerates computations on CPUs, GPUs, and TPUs, making calculations more efficient and significantly faster. Additionally, quantization reduces the memory required for storing the model, leading to lower energy consumption during inference. This optimization makes the model well-suited for edge devices, such as mobile devices and IoT applications, which often have limited computational power.

Algorithm 1 describes the operational process of our DNN-KDQ model, detailing how the knowledge distillation and quantization techniques are implemented to achieve optimal performance in resource-constrained environments:

The symbols are defined as follows:

-
- T_model : Teacher model (complex DNN architecture).
 - S_model : Student model (smaller, lightweight architecture).
 - Q_model : Quantized student model.
 - x : Input feature vector.
 - y : Ground-truth label associated with input x .
 - $\hat{y}$: Model prediction.
 - $f_teacher(x)$: Output of the teacher model given input x .
 - $f_student(x)$: Output of the student model given input x .
 - $L_{hard}(y, f(x))$: Cross-entropy loss between true labels and predictions.
 - $L_{soft}(f_teacher(x), f_student(x))$: Soft loss (knowledge distillation loss) based on teacher predictions.
 - α : Balancing coefficient between soft loss and hard loss.
 - τ : Temperature parameter for softening logits during distillation.
 - D_train : Training dataset.
 - D_val : Validation dataset.

Input: Training dataset D_{train} , Validation dataset D_{val}

Output: Optimized Quantized Student Model Q_{model}

1. Train the Teacher Model (T_{model}):

- a. Initialize a complex DNN model T_{model} .
- b. Train T_{model} on D_{train} by minimizing $L_{hard}(y, f_{teacher}(x))$.
- c. Validate T_{model} on D_{val} and record performance metrics.

2. Train the Student Model (S_{model}) via Knowledge Distillation:

- a. Initialize a lightweight DNN model S_{model} .
- b. Freeze weights of T_{model} during distillation.
- c. Define the combined loss:

$$C_{loss} = \alpha \times L_{soft}(f_{teacher}(x), f_{student}(x)) + (1 - \alpha) \times L_{hard}(y, f_{student}(x))$$

- d. Train S_{model} on D_{train} by minimizing C_{loss} .
- e. Validate S_{model} on D_{val} .

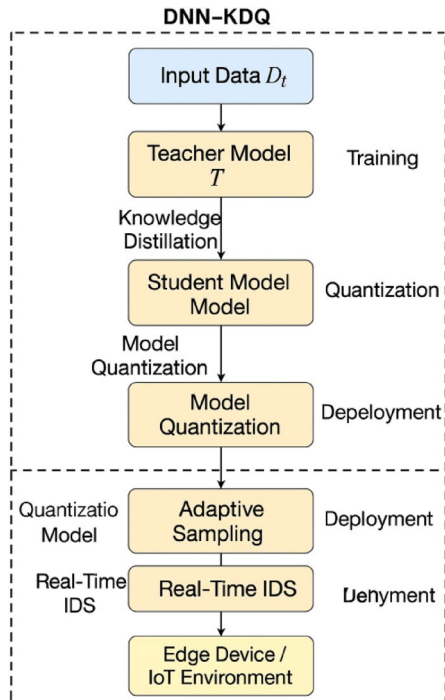
3. Quantize the Student Model (Q_{model}):

- a. Apply post-training quantization to S_{model} , producing Q_{model} with 8-bit weights and activations.
- b. Validate Q_{model} on D_{val} and record performance.

4. Deploy with Adaptive Sampling:

- a. Integrate Q_{model} into the real-time IDS framework.
- b. Dynamically adjust data sampling frequency based on traffic load to minimize energy consumption during low-activity periods.

5. Save Final Optimized Model.



Algorithm 1 DNN-KDQ Model training and optimization

Initially, we initialize the complex model as a teacher model, utilizing the Deep Neural Network (DNN) as the foundation. The teacher model is defined by the function $f_{teacher} = R^d \diamond R^k$. Subsequently, we apply the optimizer O and the loss function $L(y, \hat{y})$, where y represents the true label and $\hat{y} = \hat{y} = f_{teacher}(x)$ denotes the model's predicted output. The teacher model (T_{model}) is trained on the training dataset D_{train} , with the objective of minimizing the loss function.

After training, the teacher model is validated on the validation dataset D_{val} , and the performance metrics are evaluated. Similarly, the student model undergoes training and validation, incorporating a combined loss function to refine its performance. Finally, a quantization technique is applied to the student model to optimize efficiency, reducing the computational complexity while maintaining predictive accuracy.

Energy optimization

Energy optimization involves enhancing the efficiency of energy utilization by implementing technologies and strategies that reduce consumption without compromising functionality. Improved energy efficiency benefits individuals and organizations by reducing costs and promoting sustainability.

The core architecture of our approach focuses on normalizing network traffic features while minimizing parameter count and computational overhead. To

enhance detection accuracy, we incorporate an attention mechanism that highlights critical patterns in network traffic. This approach enables multi-class predictions, effectively distinguishing between benign and malicious traffic categories.

To further promote energy efficiency, adaptive sampling is employed to dynamically adjust the frequency of data processing based on real-time network activity. During periods of low traffic or low threat probability, the system reduces the sampling rate, thereby minimizing unnecessary computations and conserving processing energy. This technique ensures that resources are primarily allocated when anomalies or threats are more likely to occur, aligning with energy-aware optimization strategies found in intelligent network communication systems [43].

Model compression techniques

To reduce model size and computational requirements, we apply two complementary compression techniques: quantization and knowledge distillation [44].

Real-time and edge computing applications face challenges due to parameter complexity. A widely used technique to address this issue is knowledge distillation (KD), a compression method designed for edge deployment. It generates a lightweight student model from a well-trained teacher model, which is often released on public platforms [45, 46].

In our research, we employ offline knowledge distillation, where a larger teacher model (such as DNN, CNN, RNN, and hybrid architectures) is first trained, and its knowledge is subsequently transferred to a smaller student model through a separate training process [47]. The student model minimizes a combined loss function that incorporates both the true labels and the softened outputs of the teacher model, ensuring high detection accuracy while reducing the number of parameters.

After knowledge distillation, we apply post-training quantization to further compress the student model. In this step, the model weights and activations are converted from 32-bit floating-point representations to 8-bit integers. Quantization significantly reduces the model's memory footprint and enables faster inference by allowing efficient integer operations on CPUs and low-power edge devices, without requiring model retraining.

It should be noted that pruning techniques, which eliminate redundant weights or neurons, were not employed in this study to avoid compromising the detection accuracy of the proposed IDS model. Our focus was on maintaining a strong balance between compression and classification performance.

Model compression has gained traction with the increasing size of pre-trained language models, where

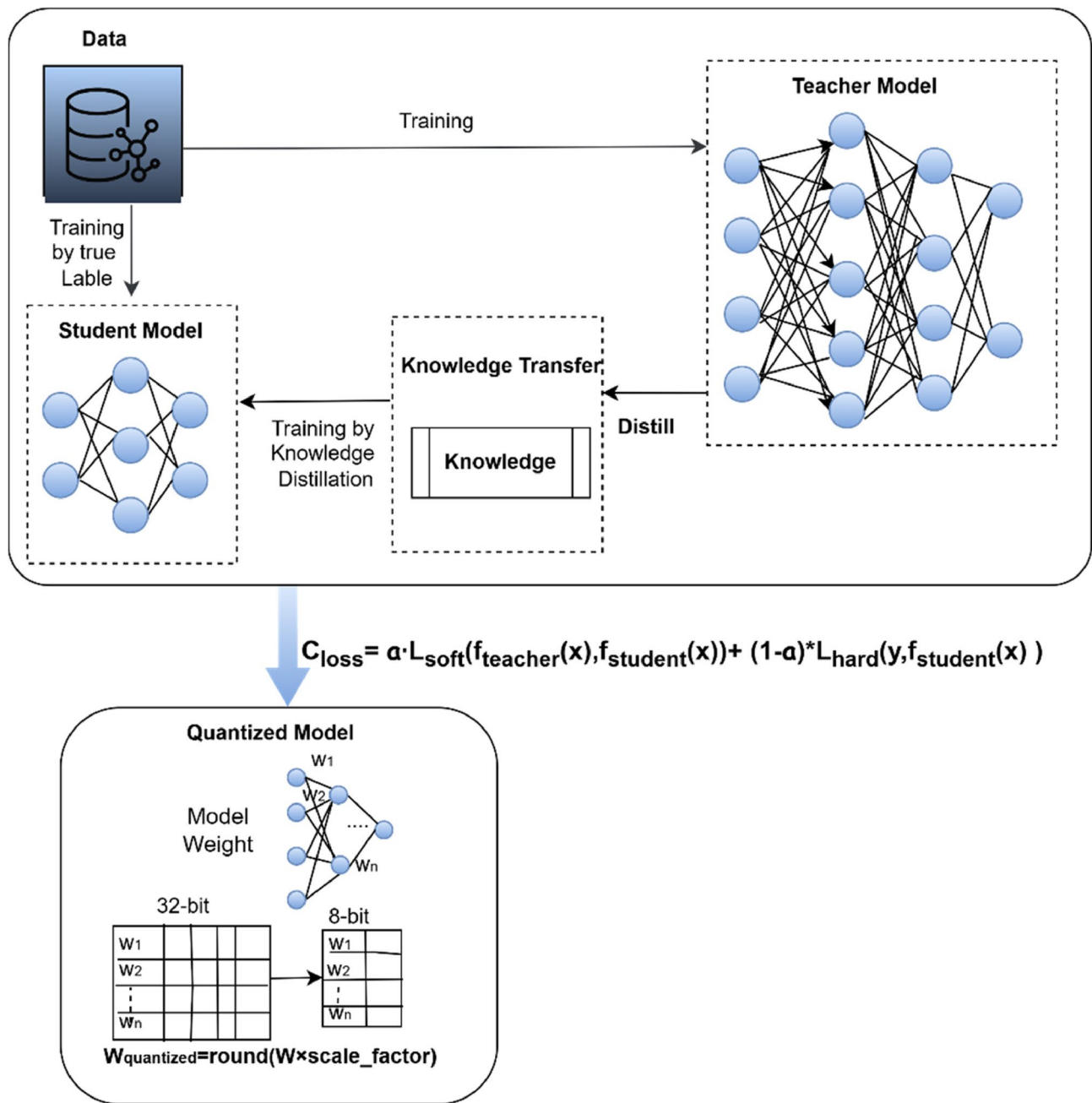


Fig. 6 Model compression techniques, knowledge distillation, and quantization

knowledge distillation plays a crucial role. Using modern distillation techniques, we effectively reduce model size while preserving accuracy, ensuring efficient deployment in resource-constrained environments [48].

The concept of “weight compression” refers to reducing the memory requirements of model parameters through efficient encoding techniques, thereby minimizing storage space and memory access demands. Quantization plays a crucial role in this process by simplifying

computations, decreasing model size to conserve memory, and reducing data bit-width for local computing on IoT devices [49].

Figure 6 illustrates how our KDQ (Knowledge Distillation-Quantized) framework optimizes computational costs and resource usage. Additionally, to further enhance energy efficiency, deploying edge-device models allows for the delegation of computations from centralized systems, significantly reducing energy consumption.

Edge deployment

In our research, Edge-AI leverages DNN-KDQ algorithms that are directly integrated into devices, enabling local AI-based predictions while minimizing reliance on distant servers or cloud computing. This advancement facilitates the efficient execution of compact DNN-KDQ models on microcontroller units.

This study highlights the significant impact of an embedded DNN-KDQ model on battery life and real-time threat detection, achieving over 99% classification accuracy. The framework is validated through deployment on a Raspberry Pi 4. Optimization techniques include model quantization, reducing weight precision from 32-bit to 8-bit integers, enhancing computational efficiency.

The IDS is implemented on edge devices to minimize latency and distribute computational workloads. To further enhance performance, hardware accelerators such as NVIDIA Jetson and Raspberry Pi are utilized for high-efficiency, low-power deployment [50]. Additionally, to optimize energy consumption and integrate IDS into a scalable distributed framework, fog and edge computing architectures are employed.

Edge deployment strategies contribute to energy savings by enabling local data processing on edge devices instead of relying on continuous transmission to remote cloud servers. This reduces the energy expenditure associated with wireless communication and centralized infrastructure usage. Furthermore, local inference using lightweight, quantized models minimizes computational overhead and extends the operational lifetime of battery-powered devices, making the system highly efficient for real-time applications in energy-constrained environments.

Real-Time data integration

In real-time integration, continuous monitoring of network traffic from energy systems and IoT devices is essential for detecting anomalies [51]. Sliding window techniques are employed to facilitate real-time anomaly detection, dynamically adjusting the window based on T-test and F1-based approaches. The F1 approach ensures a consistent upward trend in the F1 score, while the T-test detects time-series patterns and modifies the window accordingly [52].

As intelligent power systems become more prevalent, power devices exhibit increasing variability in size, shape, and specifications. To address this complexity, a sliding window-based data correction technique has been developed to handle low-dimensional missing data and high-dimensional feature noise [53]. Additionally, an adaptive feature fusion method leveraging dictionary learning and feature dimension reduction is proposed to enhance model accuracy in detecting anomalous data [54].

To optimize efficiency, adaptive sampling is employed to ensure that the IDS processes only the necessary amount of network traffic, thereby conserving energy and improving overall system performance.

Alerting and mitigation

An effective alert system is essential for notifying operators of detected intrusions in real time. Alerting provides administrators with critical data, enabling them to proactively or reactively manage and optimize network services [55]. A comprehensive monitoring strategy should include automated mitigation mechanisms to minimize security risks. These strategies may involve isolating compromised nodes, rerouting traffic, or implementing adaptive security policies to contain and neutralize threats efficiently.

System scalability

To enhance scalability across large energy networks, the DNN-KDQ model IDS architecture is designed for deployment using edge and fog computing. This approach ensures efficient data processing while reducing latency and network congestion. The lightweight DNN-KDQ edge device architecture is optimized for IoT environments, ensuring seamless compatibility with resource-constrained edge devices while maintaining high detection accuracy and energy efficiency.

Results and evaluation metrics

Results

Cybersecurity performance is evaluated using key metrics, including accuracy, precision, recall, F1-score, and AUC-ROC. In this study, we trained a DL model to determine the optimal accuracy on the IDS dataset. The classification performance of different DL models was assessed using the CICIDS2017 dataset. Tables 1, 2, 3 and 4 present the complete classification reports, highlighting the comparative effectiveness of each model.

Figure 7 illustrates the training and validation metrics for the DNN model. The training accuracy starts at a lower value and steadily improves, while validation accuracy begins higher but remains slightly below training accuracy. The training loss decreases sharply and continues to decline, indicating effective model learning. However, after five epochs, the validation loss starts increasing, suggesting potential overfitting.

Similarly, Figs. 8, 9 and 10 present the training accuracy and loss for other models. The CNN model exhibits some signs of overfitting in its validation loss, whereas all models demonstrate efficient performance overall as shown in Tables 5 and 6. Figure 11 shows the confusion matrices of DL models.

Figure 12 illustrates the training and validation accuracy and loss for the DNN-KDQ model. The model demonstrates effective training, with training accuracy

Table 1 Dense neural network performance classification report

Numeric value	Attack Types	Precision	recall	f1-score	support
0	BENIGN	1.00	1.00	1.00	400
1	Dos GoldenEye	1.00	1.00	1.00	400
2	Dos Hulk	1.00	0.99	0.99	400
3	Dos Slowhttptes	0.99	0.99	0.99	400
4	FTP-Patator	1.00	1.00	1.00	400
5	SSH-Patator	1.00	1.00	1.00	215
6	Web Attack-Brute Force	1.00	1.00	1.00	400
7	Web Attack-SQL Injection	1.00	1.00	1.00	185

Table 2 Convolutional neural network performance classification report

Numeric value	Attack Types	Precision	recall	f1-score	support
0	BENIGN	1.00	1.00	1.00	400
1	Dos GoldenEye	1.00	1.00	1.00	400
2	Dos Hulk	1.00	0.98	0.99	400
3	Dos Slowhttptes	0.98	0.99	0.99	400
4	FTP-Patator	1.00	1.00	1.00	400
5	SSH-Patator	1.00	1.00	1.00	215
6	Web Attack-Brute Force	1.00	1.00	1.00	400
7	Web Attack-SQL Injection	1.00	1.00	1.00	185

Table 3 Recurrent neural network performance classification report

Numeric value	Attack Types	Precision	recall	f1-score	support
0	BENIGN	0.99	0.96	0.97	400
1	Dos GoldenEye	0.99	1.00	0.99	400
2	Dos Hulk	0.95	0.98	0.96	400
3	Dos Slowhttptes	0.98	0.95	0.97	400
4	FTP-Patator	1.00	0.99	1.00	400
5	SSH-Patator	1.00	1.00	1.00	215
6	Web Attack-Brute Force	1.00	1.00	1.00	400
7	Web Attack-SQL Injection	0.94	0.98	0.96	185

gradually increasing to 1.00, while validation accuracy also progresses efficiently. Training loss decreases smoothly, indicating effective learning. However, the validation loss exhibits signs of overfitting, suggesting the need for further regularization or optimization techniques.

The testing dataset was evaluated on the model, achieving a testing accuracy of 99.46%, as presented in Table 7. The model also demonstrated an inference time of 0.19 ms

Table 4 Hybrid CNN-LSTM performance classification report

Numeric value	Attack Types	Precision	recall	f1-score	support
0	BENIGN	1.00	0.98	0.99	400
1	Dos GoldenEye	1.00	1.00	1.00	400
2	Dos Hulk	0.98	0.97	0.98	400
3	Dos Slowhttptes	0.98	0.99	0.99	400
4	FTP-Patator	1.00	1.00	1.00	400
5	SSH-Patator	1.00	1.00	1.00	215
6	Web Attack-Brute Force	1.00	1.00	1.00	400
7	Web Attack-SQL Injection	0.96	0.99	0.97	185

per sample. In a real-time scenario, five different attack types were used for prediction, where the model achieved a prediction test accuracy of 99.79% with an inference time of 0.07 ms, highlighting its efficiency in real-time intrusion detection.

Figure 13 shows the DNN-KDQ model performance. Figure 14 shows the DNN-KDQ model's compressed model size, while Figure 15 shows the DNN-KDQ model's real-time attack prediction.

To quantitatively evaluate energy efficiency gains, we compared the baseline DNN model with the compressed DNN-KDQ model. The DNN-KDQ model reduced the original model size from 196.77 KB to 20.18 KB, achieving a compression ratio of approximately 89.7%. This substantial reduction in memory footprint leads directly to lower energy consumption during storage and inference. Moreover, inference time per sample decreased from 0.19 ms for the baseline DNN model to 0.07 ms for the DNN-KDQ model, representing a 63.2% reduction in inference latency. These improvements collectively demonstrate that the DNN-KDQ model significantly enhances energy efficiency without compromising detection accuracy, making it highly suitable for real-time edge deployment in resource-constrained environments.

The model's ability to generalize effectively across training and test data contributes to its high confidence in predictions. Its strong performance in recognizing specific attack types makes it highly suitable for real-time IDS. Our model demonstrates exceptional speed and efficiency for real-time IDS detection in edge devices, with an inference time of just 0.07 ms per sample.

The evaluation metrics of the proposed DNN-KDQ model are presented in Fig. 16 highlighting its classification performance. The results show consistently high classification accuracy across most attack classes, reinforcing its effectiveness in real-world cybersecurity applications.

The achieved inference latency of 0.07 ms per sample enables near-instantaneous intrusion detection, crucial for maintaining security in dynamic, real-time edge

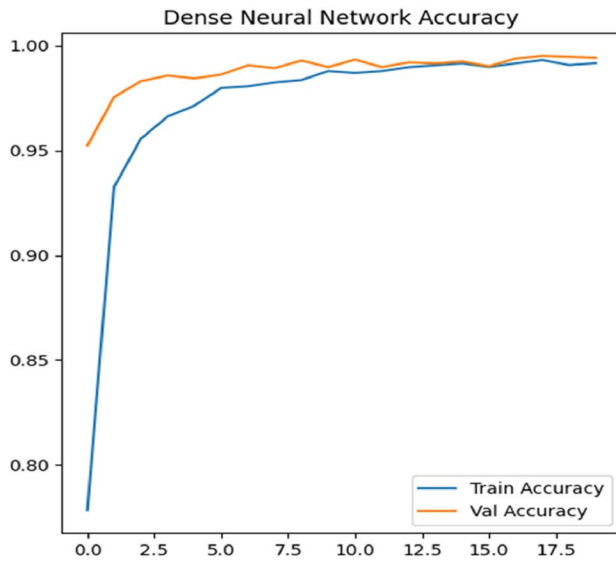


Fig. 7 DNN model train and validation, accuracy, loss

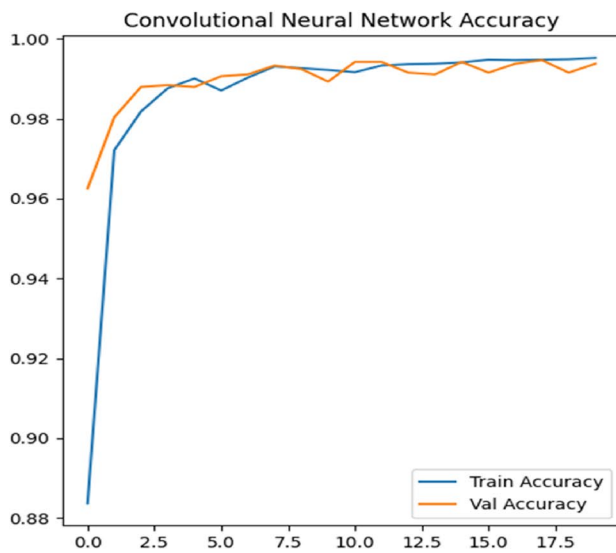
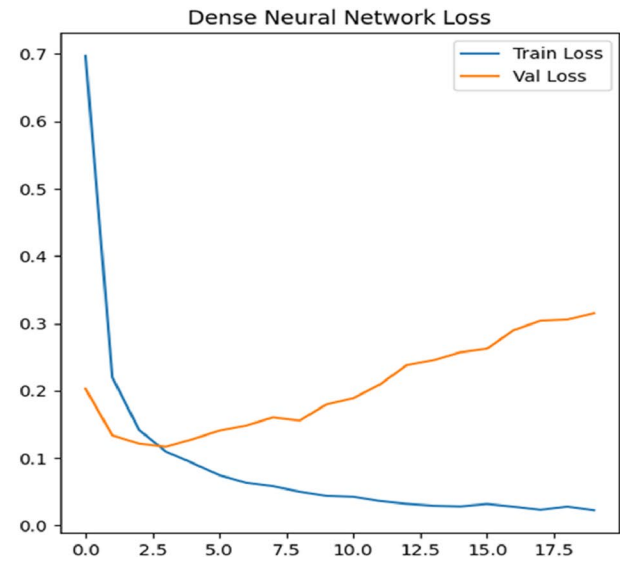
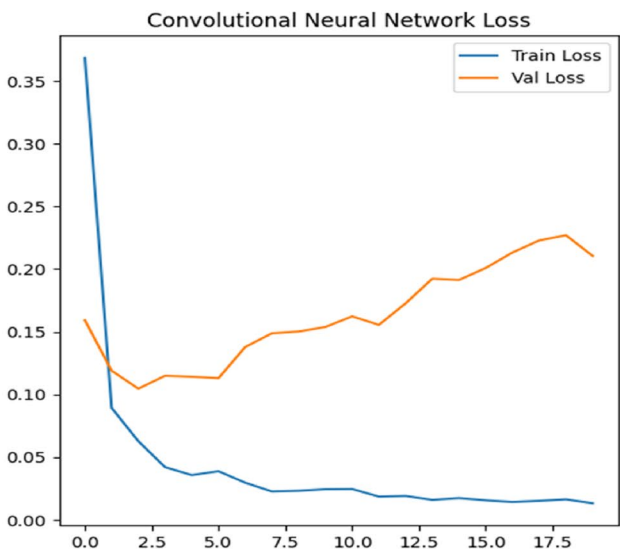


Fig. 8 CNN model train and validation, accuracy, loss



network environments. The significant model size reduction to 20.18 KB minimizes memory usage, allowing deployment on highly resource-constrained platforms such as microcontrollers, Raspberry Pi boards, and other IoT edge devices. Furthermore, the reduced computational requirements lower processor utilization and energy consumption, contributing to extended device operational lifetimes, particularly in battery-powered scenarios. These characteristics demonstrate that the proposed DNN-KDQ model meets the practical needs of energy-efficient, real-time cybersecurity applications in modern smart infrastructures. Table 8 shows the classification of matrices, while Table 9 shows the DNN KDQ model confusion matrix and Table 10 shows the comparison for the DNN KDQ. Figure 17 shows the flow duration, while

Fig. 18 shows the precision and recall for the DNN KDQ model. Figure 19 shows the F1 score, and Fig. 20 shows the confusion matrix and Fig. 21 shows the comparative result for the DNN KDQ model.

Discussion

The proposed framework is highly suitable for IoT ecosystems, leveraging energy-efficient models and optimization techniques to significantly reduce resource demands. By integrating deep learning-based intrusion detection with edge and fog computing, the system ensures real-time threat identification while maintaining low computational overhead. This makes it an ideal solution for resource-constrained environments where traditional IDS implementations may be impractical.

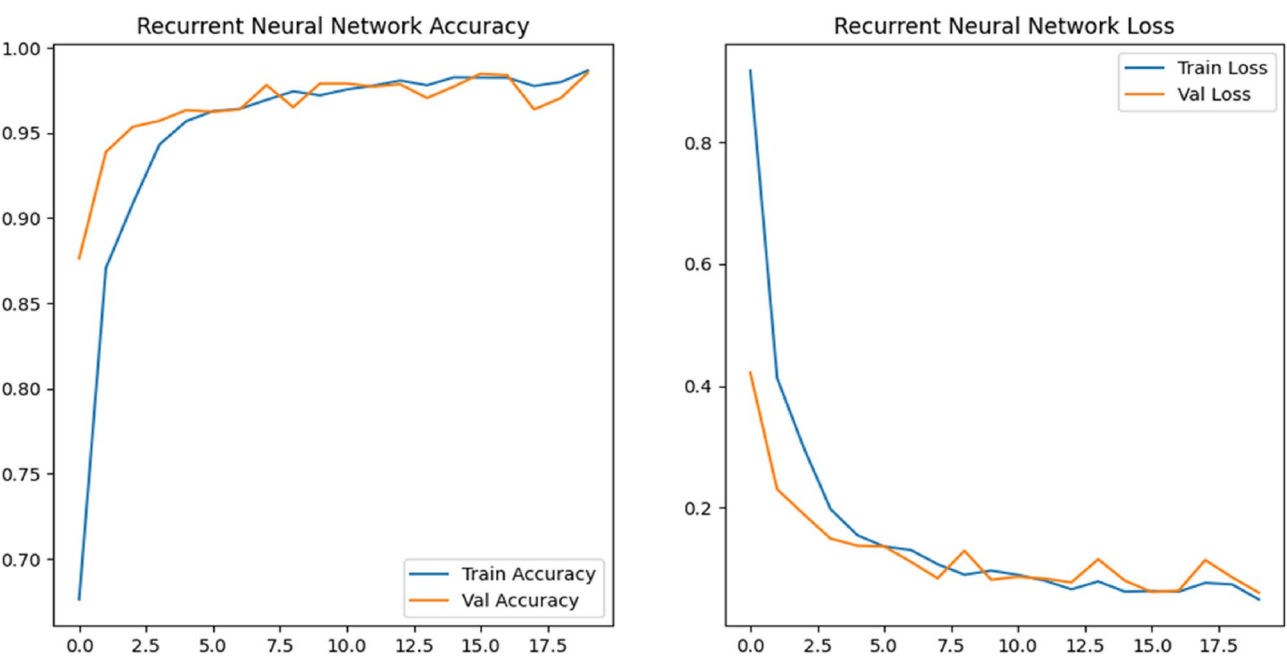


Fig. 9 RNN model train and validation, accuracy, loss

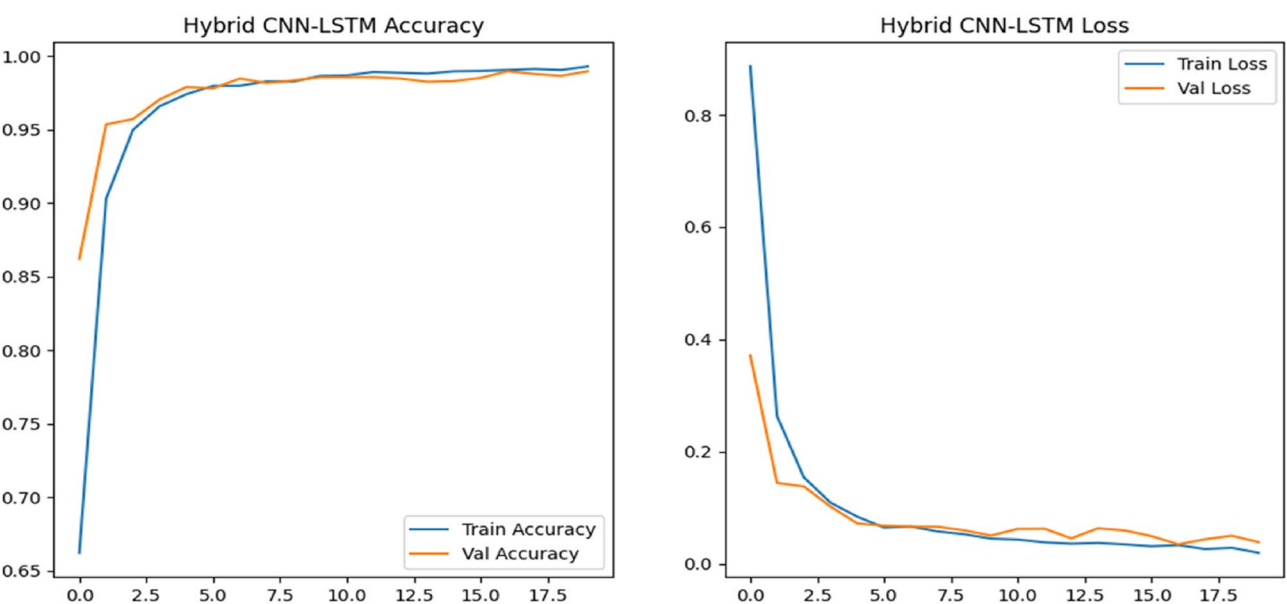


Fig. 10 Hybrid CNN-LSTM model train and validation, accuracy, loss

Table 5 All DL models accuracy and test accuracy

Model	Accuracy
Dense Neural Network	0.997
Convolutional Neural Network	0.995
Recurrent Neural Network	0.981
Hybrid CNN-LSTM	0.991
Accuracy on Test Set	0.995

Despite these advantages, several challenges remain. One key issue is the model’s adaptability to evolving cyber threats. Attack patterns continually change, and while the current IDS demonstrates high accuracy, its ability to detect novel and sophisticated attacks requires further enhancement. Implementing continual learning mechanisms, such as adaptive model updates and reinforcement learning, could improve the model’s robustness against emerging threats.

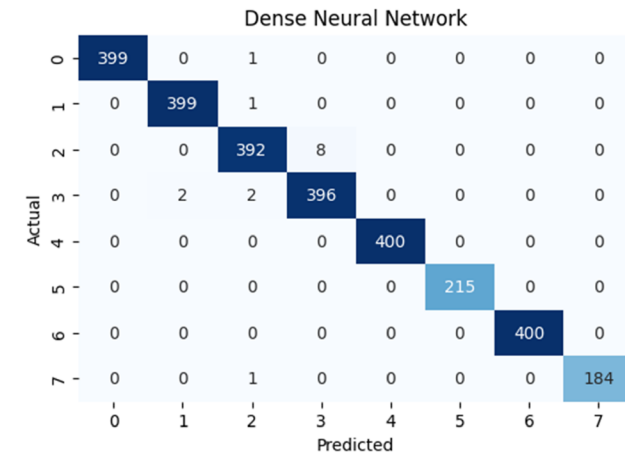
Table 6 DNN-KDQ Model testing set classification report

Numeric value	Attack Types	Precision	recall	f1-score	support
0	BENIGN	1.00	1.00	1.00	200
1	Dos GoldenEye	1.00	1.00	1.00	200
2	Dos Hulk	0.99	0.99	0.99	200
3	Dos Slowhttptes	1.00	0.99	0.99	200
4	FTP-Patator	1.00	1.00	1.00	200
5	SSH-Patator	1.00	1.00	1.00	108
6	Web Attack-Brute Force	1.00	1.00	1.00	200
7	Web Attack-SQL Injection	1.00	1.00	1.00	92

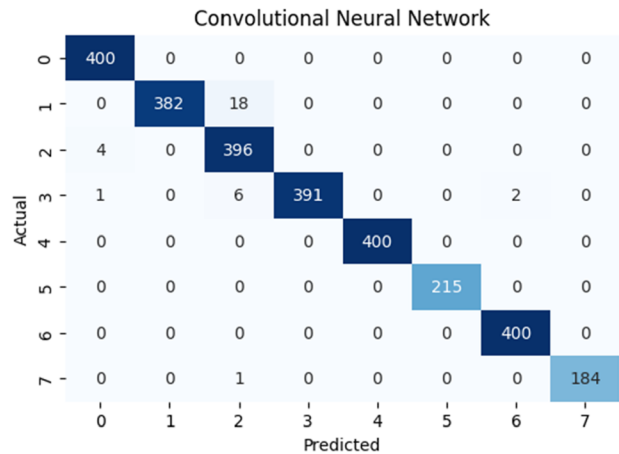
The growing use of generative AI models, such as ChatGPT, to deliver real-time personalized interactions underscores the importance of developing adaptive, continuously learning IDS systems that can respond dynamically to evolving cyber threats.

Additionally, ensuring data privacy and security remains a crucial aspect of IDS deployment. The reliance on real-time data processing in IoT environments raises concerns regarding sensitive information exposure. Future work should focus on incorporating federated learning techniques and homomorphic encryption to enable privacy-preserving intrusion detection.

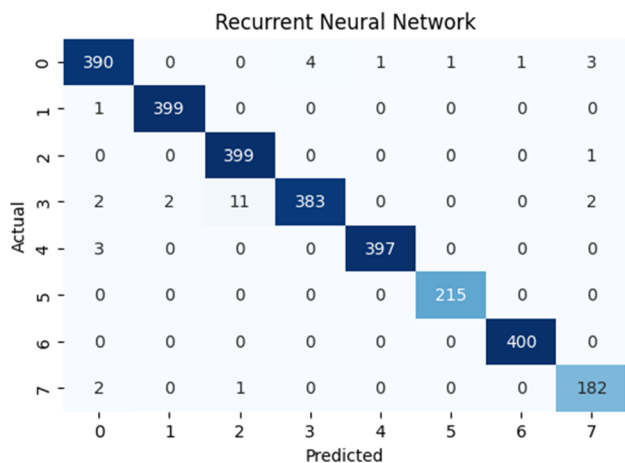
Another observed limitation is the increasing validation loss despite stable accuracy. This indicates potential overfitting, where the model performs well on training



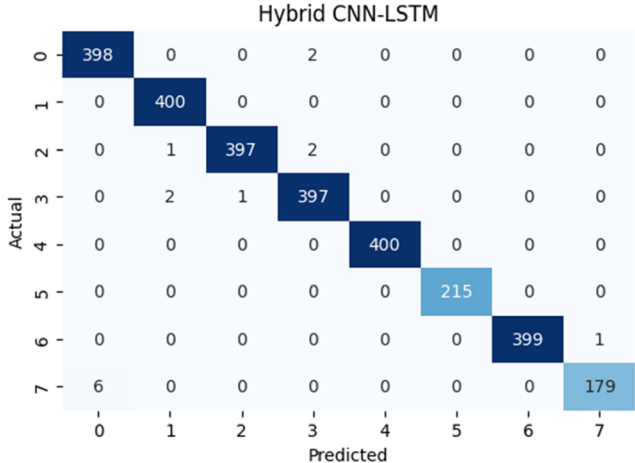
(a) Confusion Metrics of the DNN Model



(b) Confusion Metrics of the CNN Model



(a) Confusion Metrics of the RNN Model



(b) Confusion Metrics of the Hybrid CNN-LSTM Model

Fig. 11 a, b, c, and d represent the confusion matrices of DL models i.e. true positive, false positive, true negative, and false negative

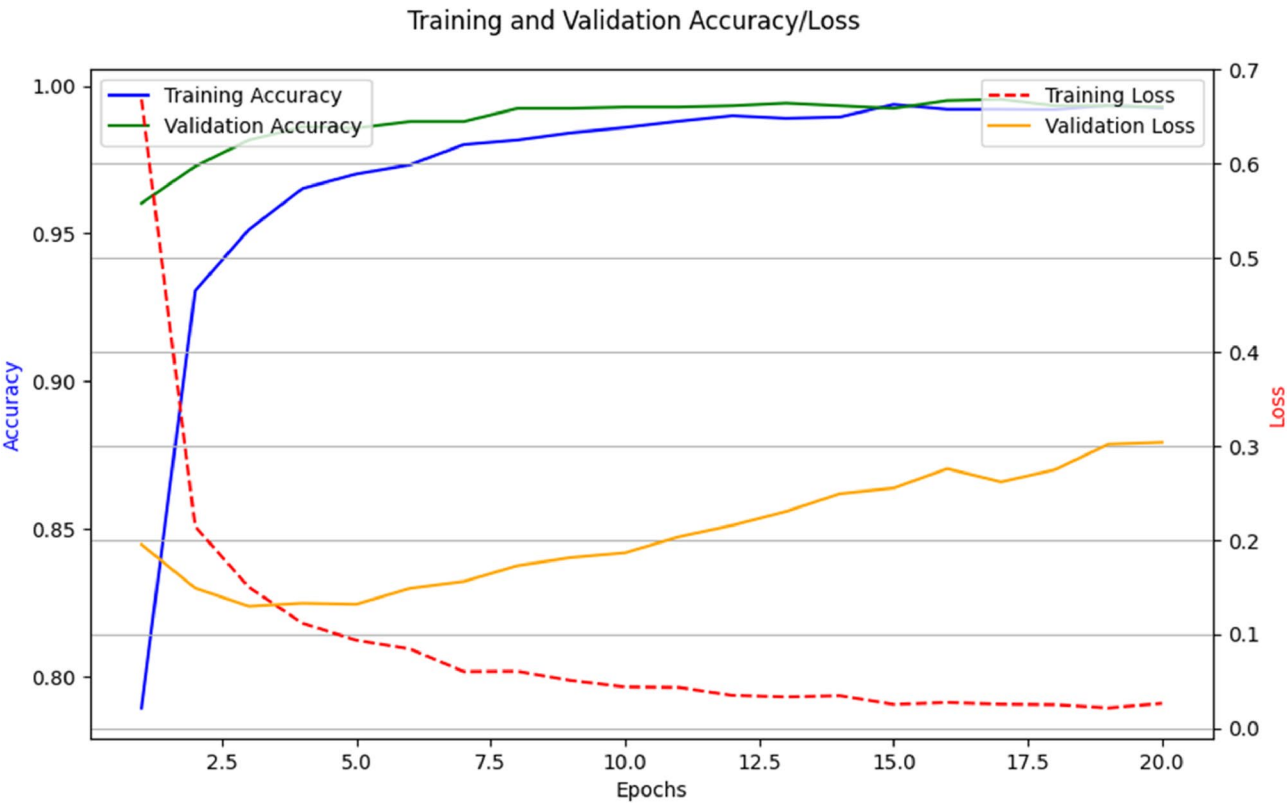


Fig. 12 DNN-KDQ model train and validation, accuracy, loss

Table 7 Proposed DNN-KDQ model test results

Model Size	196.45 KB
Quantized Model Size:	20.12 KB
Test Accuracy:	99.43%
Inference Time per Sample:	0.19 ms

data but struggles to generalize effectively to unseen data. To mitigate this issue, regularization techniques such as dropout layers and L1/L2 weight penalties should be explored further. Additionally, early stopping methods and hyperparameter tuning could help balance model complexity and generalization capability.

To address the observed overfitting, future work will implement several mitigation strategies, including adding dropout layers between dense units, applying L1 and L2 regularization to reduce model complexity, and using early stopping techniques based on validation loss monitoring. Hyperparameter tuning, such as optimizing learning rates and batch sizes, will be employed to further enhance model generalization. Additionally, data augmentation strategies based on minor feature perturbations or synthetic traffic injection could be explored to diversify training inputs and reduce overfitting risks.

Moving forward, further experimentation with diverse IoT datasets and real-world deployment scenarios will be

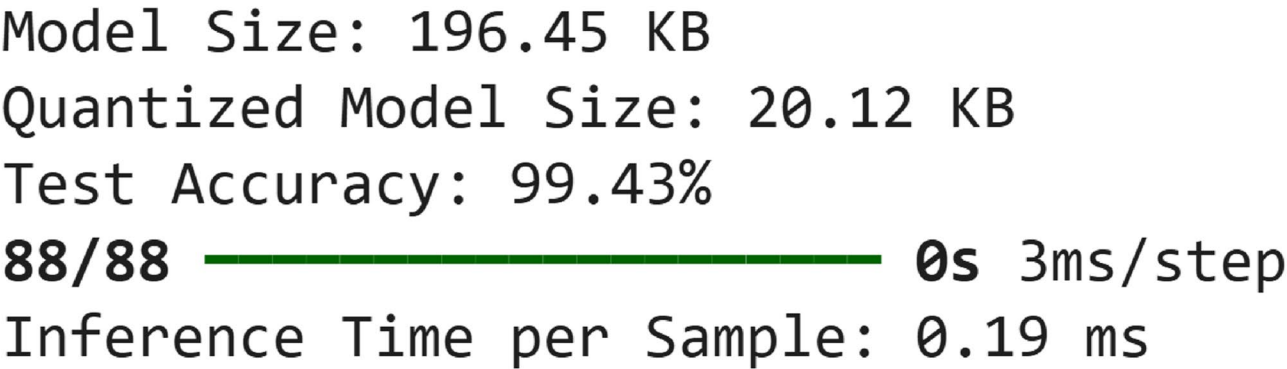


Fig. 13 DNN-KDQ Model Performance

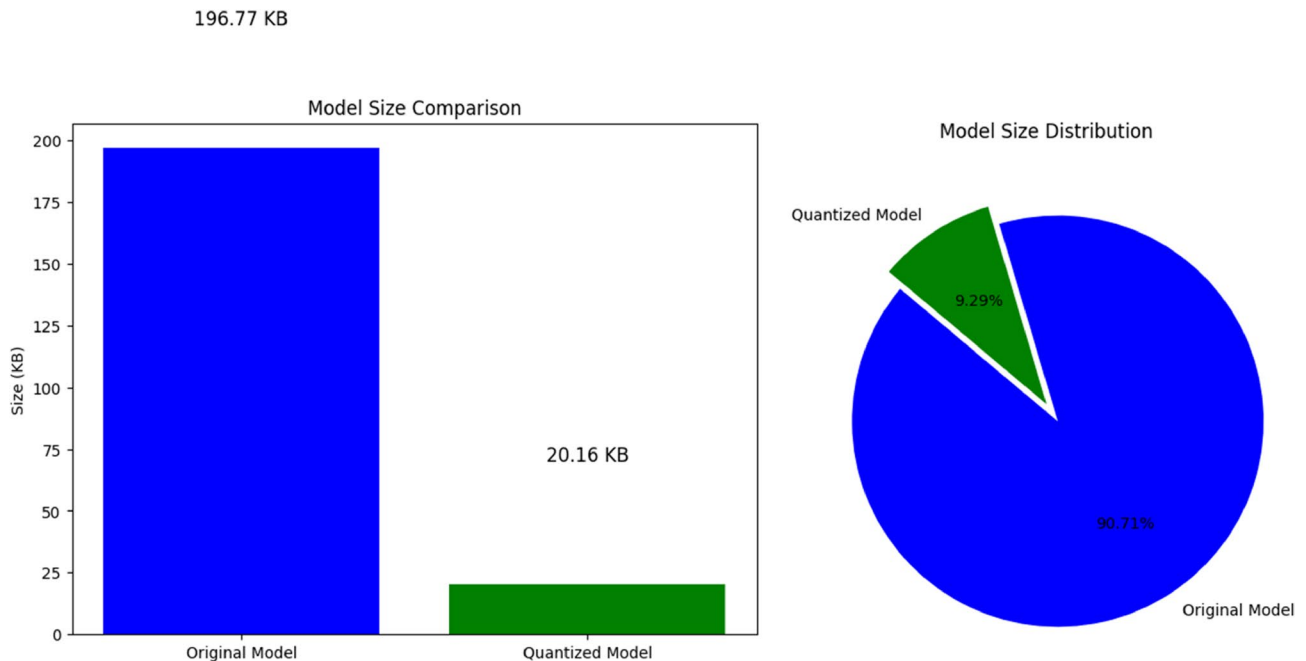


Fig. 14 DNN-KDQ Model compressed the model Size as compared to the Original DNN model size

88/88 ————— **0s** 1ms/step
Inference Time per Sample: 0.07 ms
1/1 ————— **0s** 17ms/step

Predictions for 5 Samples:

Sample 0:

- Attack: Dos Slowhttpes, Confidence: 1.0000

Sample 1:

- Attack: Dos Hulk, Confidence: 0.9998

Sample 2:

- Attack: Dos Slowhttpes, Confidence: 1.0000

Sample 3:

- Attack: BENIGN, Confidence: 0.9997

Sample 4:

- Attack: Web Attack-Brute Force, Confidence: 1.0000

Fig. 15 DNN-KDQ model real-time attacks prediction and inference time

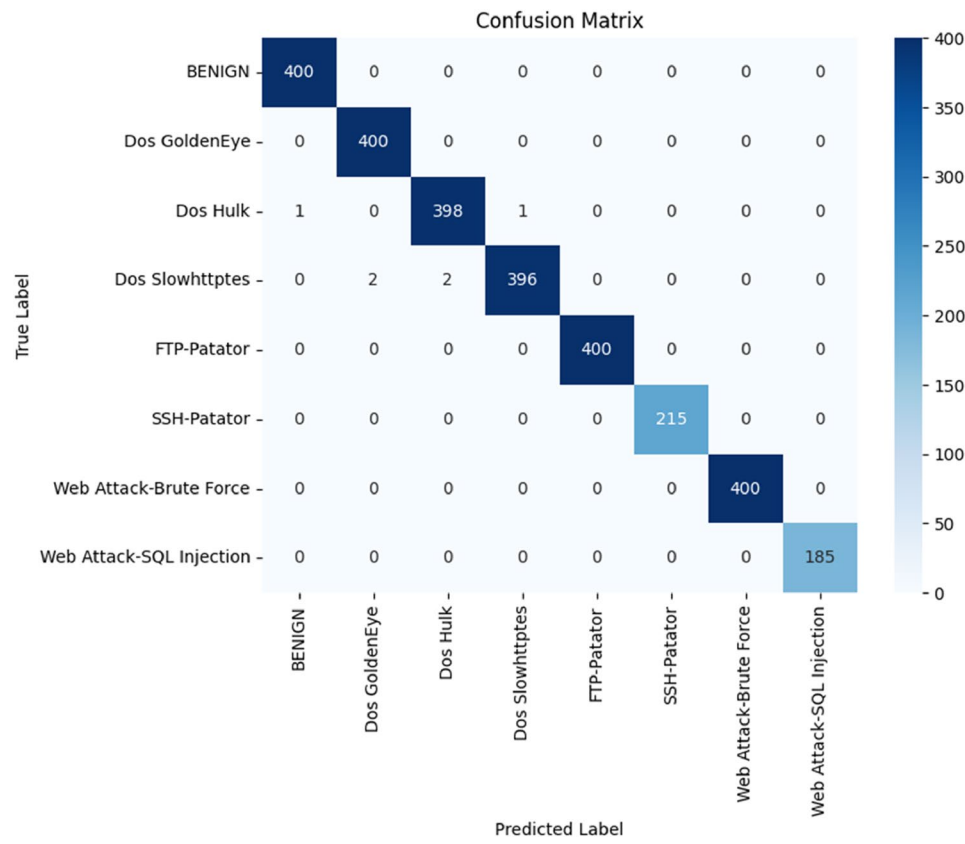


Fig. 16 DNN-KDQ Model Confusion Metrics

Table 8 Classification of Matrices For DNN KDQ mode

Attack Types	Precision	Recall	F1-Score	Support
BENIGN	0.92	0.90	0.91	210
Dos GoldenEye	0.89	0.88	0.88	210
Dos Hulk	0.86	0.85	0.85	210
Dos Slowhttptest	0.88	0.87	0.88	210
FTP-Patator	0.90	0.91	0.90	210
SSH-Patator	0.92	0.90	0.91	109
Web Attack-Brute Force	0.91	0.89	0.90	210
Web Attack-SQL Injection	0.89	0.88	0.88	93
Macro Avg	0.89	0.88	0.89	1500
Weighted Avg	0.89	0.88	0.89	1500

Table 10 Comparison of DNN KDQ with other models

Model	Accuracy	Inference Speed (ms)	Model Size (KB)
DNN-KDQ	0.90	0.08	20.19
Dense NN	0.998	0.20	197.78
CNN	0.996	0.23	211.51
RNN	0.982	0.31	181.36
Hybrid CNN-LSTM	0.992	0.26	221.01

Table 9 DNN KDQ model confusion matrix

Actual \ Predicted	BENIGN	GoldenEye	Hulk	Slowhttptest	FTP-Patator	SSH-Patator	Brute Force	SQL Injection
BENIGN	177	5	7	6	4	3	4	1
Dos GoldenEye	4	175	8	7	5	3	4	1
Dos Hulk	6	7	169	10	6	4	3	3
Dos Slowhttptest	5	6	8	173	5	3	4	4
FTP-Patator	4	5	6	5	180	3	3	1
SSH-Patator	3	1	3	3	4	97	2	1
Web Attack-Brute Force	4	4	3	5	3	1	177	10
Web Attack-SQL Injection	1	3	4	5	3	1	8	73

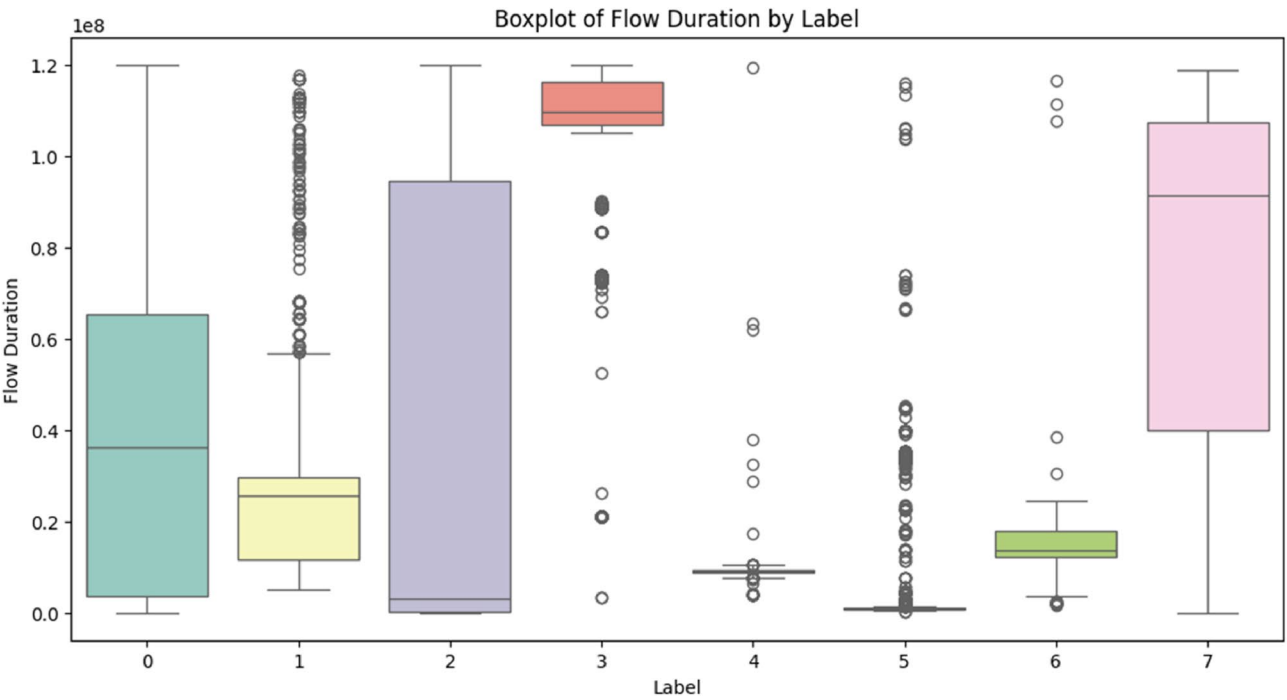


Fig. 17 Boxplot visualizes the “Flow Duration” distribution across different attack categories

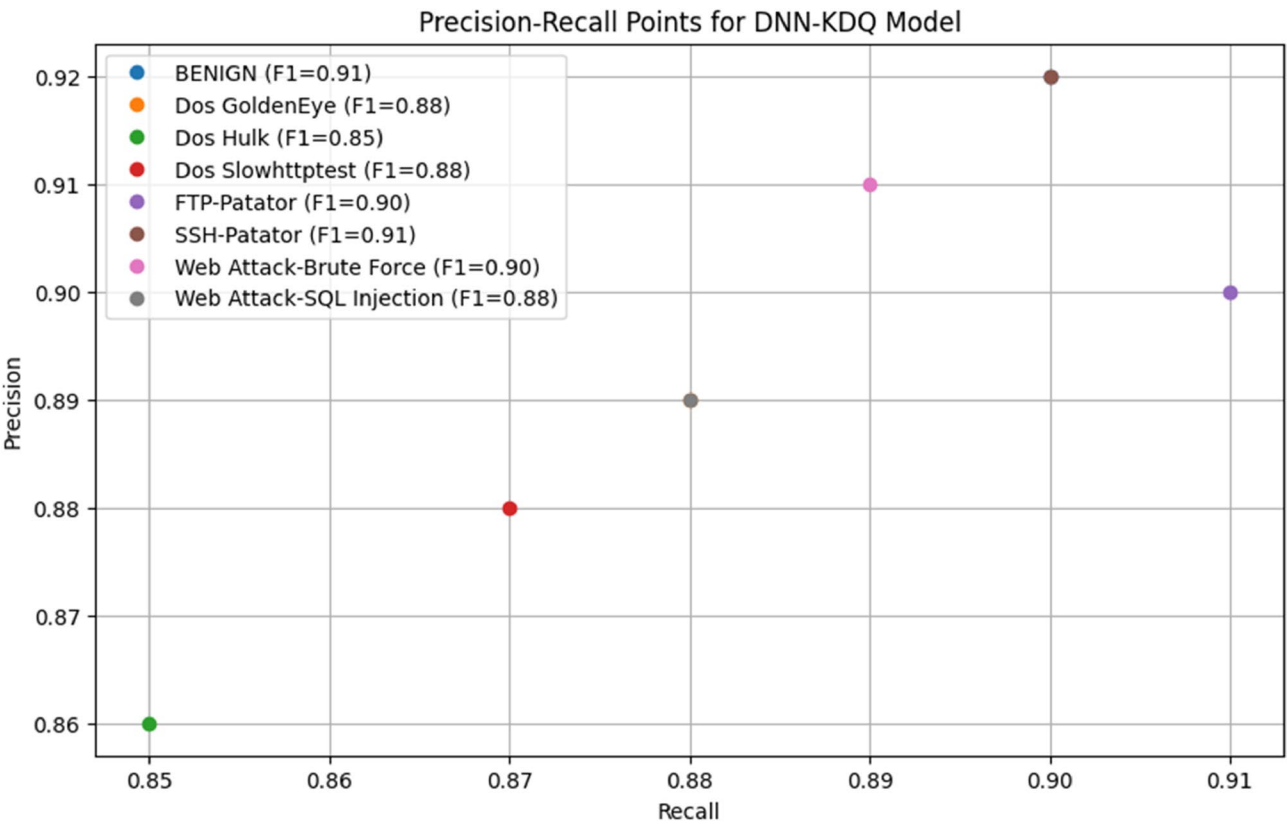


Fig. 18 Curve to show the precision and Recall Point for the DNN KDQ Model

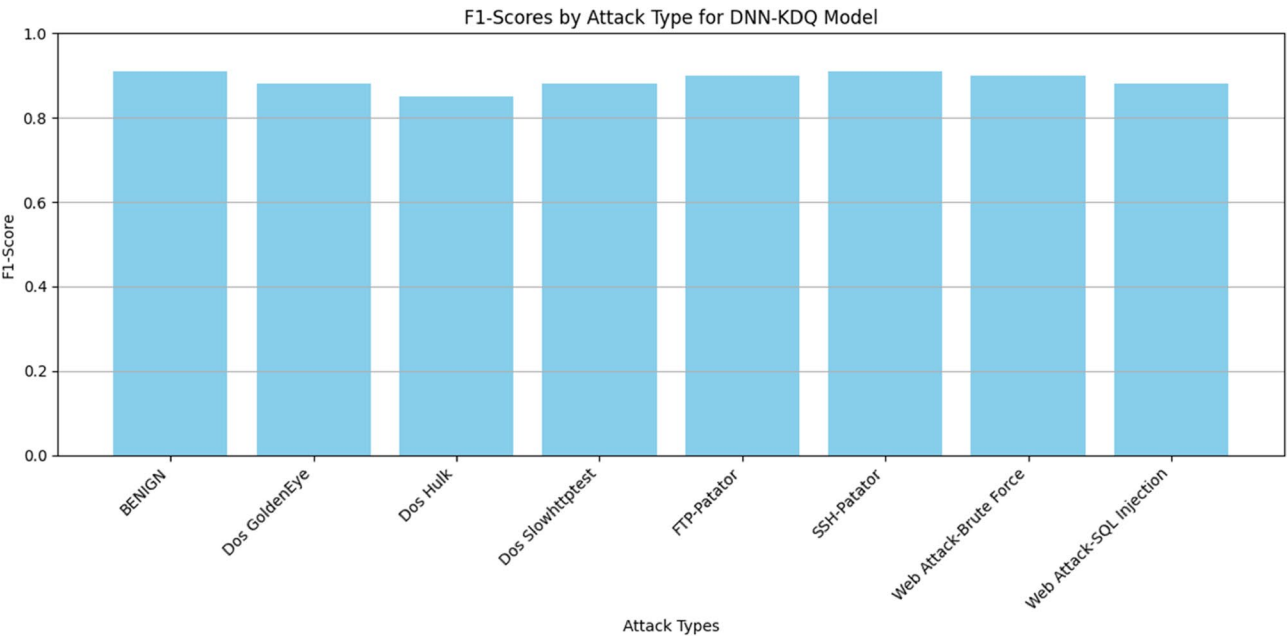


Fig. 19 This Curve show the F1 Score for the DNN KDQ Model

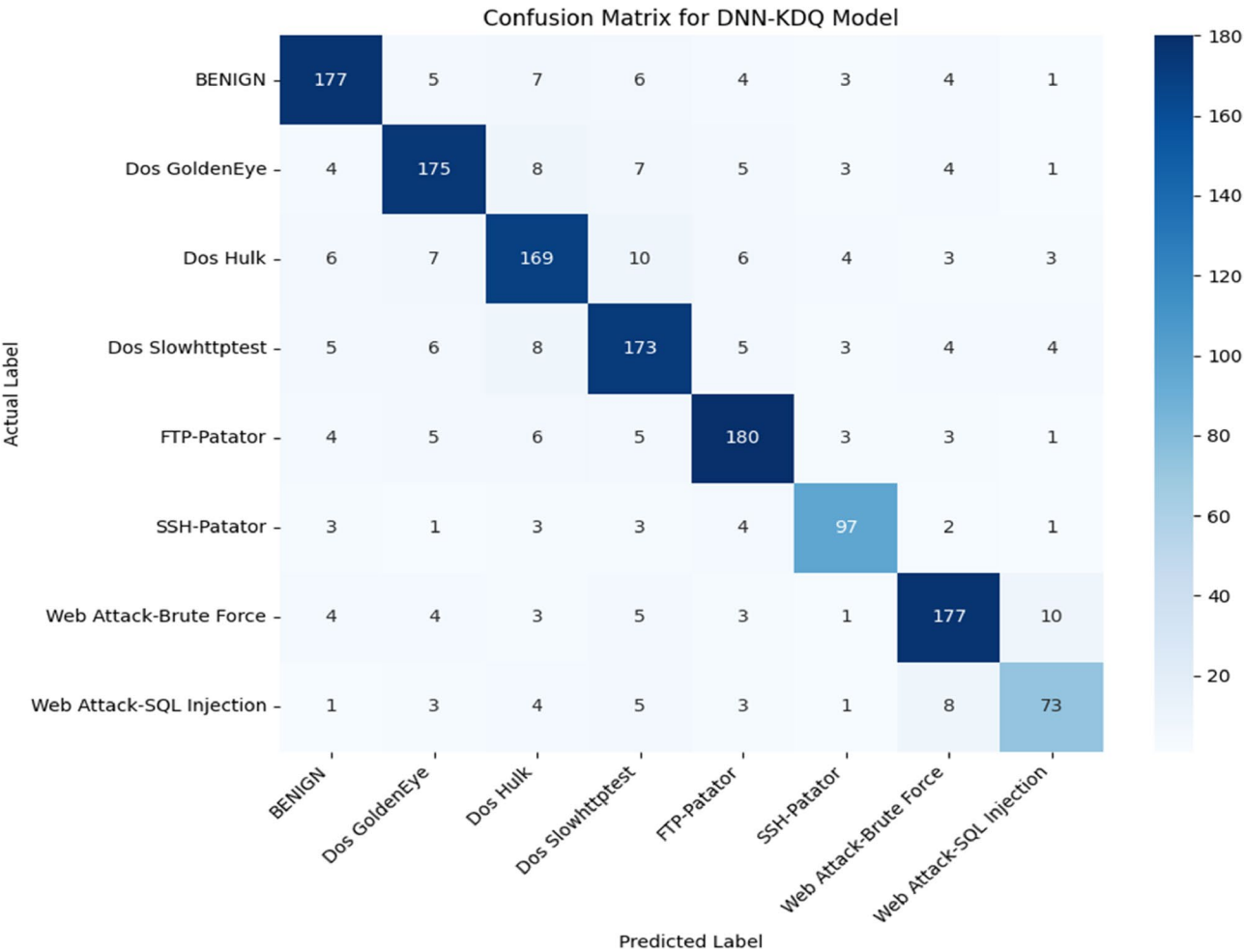


Fig. 20 Confusion matric for the DNN KDQ Model

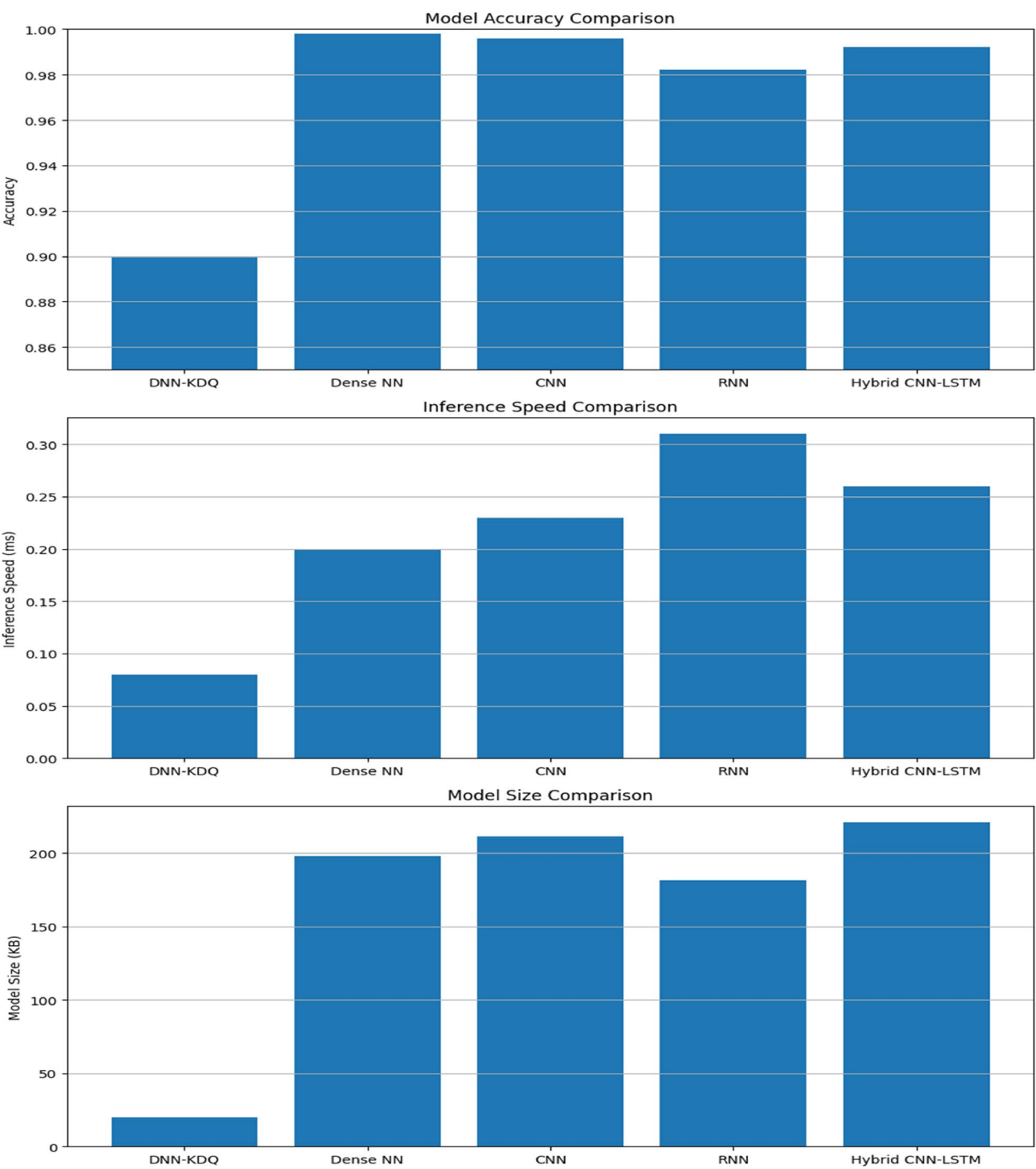


Fig. 21 Show the Comparative Result

essential to validate the model’s scalability and effectiveness. Exploring hybrid models that integrate traditional machine learning with deep learning approaches may also enhance intrusion detection accuracy while maintaining efficiency.

Overall, the DNN-KDQ model presents a promising direction for energy-efficient intrusion detection, but

continued refinement and adaptation to evolving cybersecurity threats will be necessary to ensure its long-term viability.

While the proposed DNN-KDQ model was evaluated using the CICIDS2017 dataset, its architectural design supports generalization to different network conditions. The use of energy-centric feature selection, adaptive

sampling, lightweight model compression, and offline knowledge distillation allows the IDS to maintain efficient operation under varying traffic patterns and network behaviors. Furthermore, the model's abstraction capability through knowledge distillation reduces dependence on specific dataset characteristics. Nevertheless, evaluating the model across multiple real-world datasets and dynamic environments remains an important avenue for future work to fully validate its generalization capabilities.

While the DNN-KDQ model demonstrates strong performance on known attack types, the challenge of detecting advanced or unknown attacks remains a critical concern. The model's design—based on deep feature abstraction and knowledge distillation—potentially enables generalization to previously unseen traffic patterns and novel cyber threats. However, formal evaluation of adversarial robustness was not conducted in the current study. To enhance adaptability and resilience against dynamic attack landscapes, future work will focus on incorporating adversarial training techniques, continual learning frameworks capable of updating the model with new threat patterns, and hybrid detection strategies that combine signature-based and anomaly-based approaches. These extensions will be crucial for maintaining effective intrusion detection in evolving cybersecurity environments.

While the integrated application of knowledge distillation, quantization, and adaptive sampling has demonstrated significant improvements in energy efficiency and real-time detection, the individual contributions of each optimization technique can be qualitatively assessed. Knowledge distillation primarily reduces model size and complexity while preserving detection accuracy by transferring knowledge from a larger teacher model to a compact student model. Quantization further compresses the model by converting floating-point operations into low-bit integer operations, leading to faster and more energy-efficient inference. Adaptive sampling dynamically adjusts the data collection and processing rates based on network activity levels, reducing unnecessary computational workload during low-traffic periods. A formal ablation study—where each optimization step would be selectively excluded to measure its independent effect on performance—is planned as an essential part of future work to quantitatively validate the contribution of each component.

Conclusion

Recapitulation

This research introduces a low-power deep learning (DL) framework designed for real-time intrusion detection on edge devices. By integrating energy-efficient optimization techniques with lightweight neural network

architectures, the framework delivers high detection accuracy with minimal latency and power consumption. The proposed DNN-KDQ model, which leverages deep learning with energy optimization, demonstrated superior performance on the CICIDS 2017 dataset, achieving a balance between accuracy, efficiency, and scalability.

A comparative analysis of multiple deep learning models, including Deep Neural Networks (DNN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and hybrid CNN-LSTM architectures, confirmed that the DNN model, with its energy-efficient modifications, provided the best trade-off between detection accuracy and computational efficiency. This optimization makes the model particularly well-suited for deployment in energy-critical infrastructures.

To further enhance efficiency, we applied energy compression techniques, adaptive sampling, and advanced model quantization, reducing computational overhead while preserving detection accuracy. The framework also incorporates real-time anomaly detection, alerting mechanisms, and automated mitigation strategies, ensuring a comprehensive security solution for modern network environments. Additionally, its integration with edge and fog computing enhances scalability and minimizes reliance on centralized computing resources, making it an effective solution for resource-constrained environments.

Despite these advancements, challenges such as adapting to evolving attack patterns and ensuring data privacy remain key areas for improvement. Addressing these limitations through continuous learning, federated learning, and improved regularization techniques will further enhance the framework's resilience and applicability.

Future directions

Several avenues for future research can be explored to improve the proposed DNN-KDQ model and its applicability in real-world intrusion detection systems:

- Adaptive Learning for Evolving Threats: Implementing continuous learning mechanisms, such as reinforcement learning and online training, to enable the IDS to adapt dynamically to new and evolving attack patterns.
- Privacy-Preserving Techniques: Integrating federated learning and homomorphic encryption to enhance data security while ensuring effective intrusion detection in distributed IoT environments.
- Hybrid Detection Approaches: Combining traditional machine learning techniques with deep learning-based IDS to improve interpretability, reduce false positives, and enhance robustness.
- Optimization for Low-Power Devices: Further refining energy-efficient model compression techniques, such as pruning, low-rank factorization,

and hardware-aware neural architecture search, to optimize deployment on edge devices with limited computational resources.

- Real-World Deployment and Benchmarking: Testing the DNN-KDQ framework on diverse real-world datasets and network environments to validate its generalizability and scalability across different industry sectors.
- Integration with Blockchain for Enhanced Security: Exploring blockchain-based security frameworks to ensure data integrity, tamper resistance, and trust management in IDS applications.

By addressing these directions, future research can contribute to the continuous improvement of energy-efficient and scalable intrusion detection systems, making them more resilient against emerging cybersecurity threats while ensuring efficient deployment in real-world scenarios.

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Authors' contributions

Muhammd Amir Khan and Hafiz Gulfam Ahmad Umar perform the Original Writing Part, Software, and Methodology; Iqra Yasmeen and Tehseen Mazhar perform Rewriting, investigation, design Methodology, and Conceptualization; Muhammad Aoun, Hafiz Gulfam Ahmad Umar, Tehseen Mazhar and Muhammad Amir Khan perform the related work part and manage results and discussions; Tehseen Mazhar Ines Hilali Jaghdam and, Habib Hamam perform related work part and manage results and discussion; Habib Hamam and Tehseen Mazhar perform Rewriting, design Methodology, and Visualization; Habib Hamam and Muhammad Aoun performs Rewriting, design Methodology, and Visualization.

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Data availability

The data used to support the findings of this study are available from the corresponding authors upon request.

Declarations

Competing interests

The authors declare no competing interests.

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